Precision in multiboson calculations and vector boson polarizations

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August 27, 2026

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Preamble

These lecture notes were prepared for the COMETA Multiboson School, taking place in Split from 24th August to 4th September 2026. The first pillar is a hopefully self-sufficient review of radiative corrections at hadron colliders and their implications for multi-boson production processes, especially those of polarised massive vector bosons. The second pillar is the phenomenological consequences of these corrections.

This lecture is phenomenologically oriented, meaning we will focus on the theoretical aspects most relevant to making precise predictions for LHC observables in multi-boson processes. Due to time constraints, we will not be able to dive deeply into the theoretical foundations behind many of the topics we discuss. Radiative corrections, in particular, are crucial to LHC phenomenology but also a highly technical topic. The lecture aims to review the most important aspects but may be too shallow for interested readers; we point to the existing excellent literature on QCD and related subjects, for example [1, 2, 3, 4].

1 Introduction

In the post-Higgs-discovery age, one of the main targets of particle physics remains: to look for signatures that cannot be explained within the Standard Model of Particle physics (SM). We expect the SM to be incomplete, an effective theory if one wishes so, because we have evidence for phenomena outside of particle physics, in particular astrophysics and cosmology, that go beyond the standard framework of the SM and general relativity. This includes the observed matter-antimatter imbalance, evidence for dark matter and dark energy, and the unclear nature of neutrino masses.

However, starting from the perspective that the SM is incredibly successful in describing collision data collected at the LHC (see the ATLAS SM summary in Fig. 1) and its predecessors, we have to look out for very subtle effects implied by new interactions or particles, or require exploring extreme corners of the phase space[1]. This can take the form of precision measurements of SM quantities, which are interesting for their own sake but also in order to hunt for inconsistencies, or, for example, global fits of effective field theories describing heavy new physics. Both approaches require a high-fidelity understanding

[1] Of course, if there is a new resonance directly produced in the bulk of the phase space, the whole picture would change. So far we don't see anything directly produced in bulk, and the HL-LHC, with essentially the same phase space as the LHC, will not drastically change this.

of SM backgrounds (foregrounds would be the better word).

The electroweak sector is of particular interest because it connects all parts of the SM; it offers the opportunity to study a non-abelian gauge theory that is not strongly coupled (at least for energies reachable for us, in contrast to QCD) and allows us to study the nature of the electroweak symmetry breaking, which is one of the least understood parts of the SM. There are many ways to approach this subject, and even more interesting phenomenological signatures to study. For the purpose of the lecture, we will focus on the electroweak production of boson pairs in direct production and vector boson scattering channels.

This lecture introduces precision phenomenology, with an emphasis on multi-boson production, and ends with a discussion on polarised vector bosons. Crucially, because the LHC initial state is hadronic, we will also spend substantial time on QCD, which provides the dominant source of radiative corrections and is thus indispensable for a precise description of the data. The lecture is structured as follows: we start with a review of the EW SM and EWSB in section 2, followed by a discussion of event simulations in hadron colliders and radiative corrections using QCD as an example in section 3. After this groundwork, we will dive into multi-boson phenomenology with emphasis on radiative effects in section 4. The lecture closes with a treatment of polarised states in section 5.

2 SM & Electroweak Symmetry Breaking

Before diving into radiative corrections and phenomenology, we will first review the EW part of the Standard Model. This will set the stage and also introduce some of the notations used throughout the lecture.

2.1 Gauge symmetries and vector boson masses

The Standard Model of particle physics is formulated as a Yang-Mills (YM) gauge theory governed by the symmetry group:

$$\mathcal{G}_{\text{SM}} = SU(3)_C \times SU(2)_L \times U(1)_Y. \tag{1}$$

The YM gauge Lagrangian, before electroweak symmetry breaking, reads:

$$\mathcal{L} = \bar{\Psi}_{f,a}(i\not{D} - m_f\delta_{ab})\Psi_{f,b} - \frac{1}{4}F^A_{i,\mu\nu}F_i^{A,\mu\nu} \text{ with } (D_\alpha)_{ab} = \partial_\alpha\delta_{ab} + ig_i(t^C A^C_{i,\alpha})_{ab} \tag{2}$$

$$\text{and the field strength tensor: } F^A_{i,\mu\nu} = \partial_\mu A^A_{i,\nu} - \partial_\nu A^A_{i,\mu} - g_i f_i^{ABC} A^B_{i,\mu} A^C_{i,\nu} \tag{3}$$

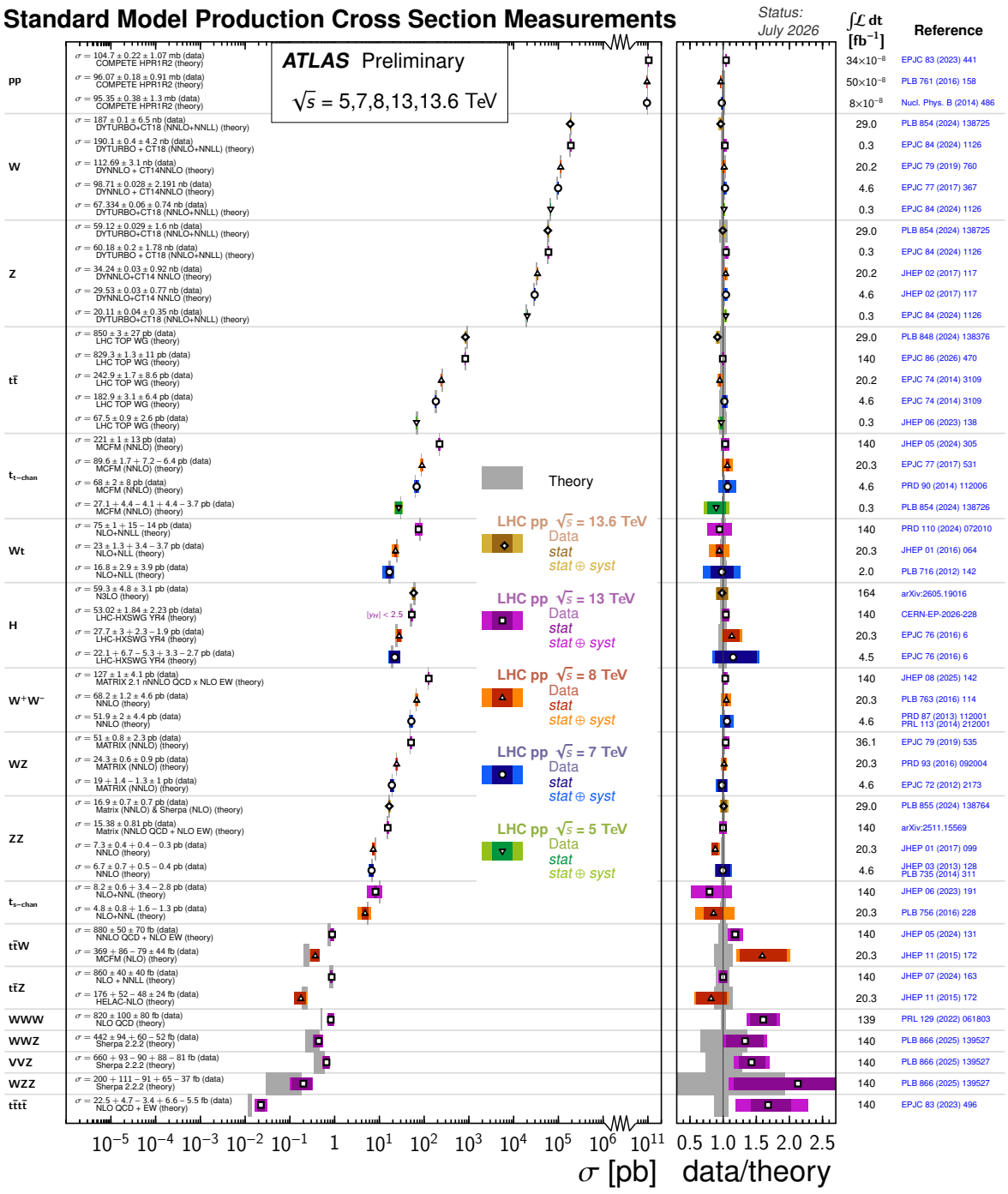


Figure 1: ATLAS SM measurement summary plot [5].

The fields $\Psi_{f,a}$ denote fermions with mass m_f. The fermion fields here are all multiplets in the respective gauge theories, denoted by the index a.

In the electroweak sector $(SU(2)_L \times U(1)_Y)$, the gauge fields comprise:

- Three weak isospin gauge fields $A^A_{SU(2)_L,\mu} = W^A_\mu$ $(A = 1, 2, 3)$ associated with generators $t^A = \sigma^A/2$ and coupling constant g.
- One weak hypercharge gauge field $A_{U(1)_Y,\mu} = B_\mu$ associated with generator $t = 1$ and coupling constant g'. (Here index A has only one value and therefore is dropped).

The non-Abelian and Abelian field-strength tensors are respectively as:

$$W^A_{\mu\nu} = \partial_\mu W^A_\nu - \partial_\nu W^A_\mu + g\epsilon^{ABC} W^A_\mu W^C_\nu, \tag{4}$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \tag{5}$$

From this you can also read off the structure constants f_i^{ABC} of the underlying $SU(2)$ $(f^{ABC} = \epsilon^{ABC})$ and $U(1)$ $(f^{ABC} = 0)$ Lie-Group. Under an infinitesimal local gauge transformation parametrized by $\alpha^a(x)$ and $\beta(x)$, the gauge bosons transform as:

$$W^a_\mu \to W^a_\mu - \frac{1}{g}\partial_\mu \alpha^a - \epsilon^{abc}\alpha^b W^c_\mu, \tag{6}$$

$$B_\mu \to B_\mu - \frac{1}{g'}\partial_\mu \beta. \tag{7}$$

All bosons here are massless but we would like to describe the observed massive $W^\pm$ and Z bosons as well as the massless photon with the $(SU(2)_L \times U(1)_Y)$. An explicit, ad-hoc mass term $\frac{1}{2}M^2 W^a_\mu W^{\mu,a}$ violates gauge invariance:

$$\frac{1}{2}M^2 W^a_\mu W^{\mu,a} \longrightarrow \frac{1}{2}M^2 W^a_\mu W^{\mu,a} - \frac{M^2}{g}(\partial_\mu \alpha^a) W^{\mu,a} + \mathcal{O}(\alpha^2) \neq \frac{1}{2}M^2 W^a_\mu W^{\mu,a}. \tag{8}$$

Such terms ruin the underlying Ward-Takahashi and Slavnov-Taylor identities, rendering the quantum field theory unrenormalizable and leading to non-unitary scattering amplitudes at high energies.

The Brout-Englert-Higgs mechanism resolves this issue. Before reviewing the BEH, let's review gauge choices.

2.2 Gauge choices

In non-Abelian gauge theories ($SU(N)$ or $SU(2)_L \times U(1)_Y$), the path integral over gauge configurations,

$$Z = \int \mathcal{D}A \, \exp\left(i \int \mathrm{d}^4x \, \mathcal{L}_{\text{inv}}[A]\right), \tag{9}$$

is ill-defined because the gauge-invariant action $\mathcal{L}_{\text{inv}}[A]$ is flat along gauge orbits generated by transformations $A_\mu \to A^\alpha_\mu$. Integrating along these non-physical, infinitely redundant orbits makes the functional measure diverge and renders the gauge-field propagator non-invertible. Another, less formal, way to see that is to look into perturbation theory. An issue can be seen when looking at the action in momentum space:

$$S_{\text{YM}} = i \int \mathrm{d}^4x \left\{ -\frac{1}{4} F^a_{\mu\nu} F^{a,\mu\nu} \right\} \ni \frac{i}{2} \int \mathrm{d}^4x A^a_\mu (\partial^2 g^{\mu\nu} - \partial^\mu \partial^\nu) A^b_\nu \delta_{ab} \tag{10}$$

In perturbation theory, i.e., the derivation of the Feynman rules, the A fields' propagator is the inverse of the bilinear form, i.e.:

$$i\Delta_{\mu\rho} \left(p^2 g^{\rho\nu} - p^\rho p^\nu \right) = g^\nu_\mu \tag{11}$$

but we can easily see:

$$i\Delta_{\mu\rho} \left(p^2 g^{\rho\nu} - p^\rho p^\nu \right) p_\nu = 0 \neq p_\mu \tag{12}$$

The main reason for this issue is that too many degrees of freedom propagate. A possibility to solve it is to use a local gauge transformation to eliminate it; this procedure is called gauge fixing.

In general, gauge fixing can be implemented through Lagrange multipliers on the Lagrangian level. Two broad classes of gauges are commonly used. The covariant gauges, or R_ξ-gauges (I will use λ or ξ as the Lagrange parameter), use the additional constraint

$$\partial \mathcal{A} = 0 \quad \to \mathcal{L}_{\text{GF}} = -\frac{1}{2\lambda} \left(\partial_\mu \mathcal{A}^\mu \right)^2 \ \lambda \in \mathbb{R} \tag{13}$$

This leads to an additional term in the bilinear form, and we can find the inverse easily:

$$i\Delta_{\mu\rho} \left(p^2 g^{\rho\nu} - (1 - \frac{1}{\lambda}) p^\rho p^\nu \right) = g^\nu_\mu \tag{14}$$

$$\Rightarrow \Delta_{\mu\nu} = \frac{-i}{p^2 + i\epsilon} \left[g_{\mu\nu} - (1 - \lambda) \frac{p^\mu p^\nu}{p^2} \right] \tag{15}$$

While this yields an invertible propagator, it also allows non-transverse degrees of freedom to propagate. The Faddeev-Popov procedure introduces ghost particles that precisely cancel these contributions (see the short discussion below). In any case, the propagator depends on the additional parameter λ, the physical results (i.e. matrix elements) do not depend on it (super useful for cross checks). Famous choices are $\lambda = 1$ (Feynman gauge) and $\lambda = 0$ (Landau gauge).

The other option are physical gauges $n_\mu A^\mu = 0$ where $n \cdot p \neq 0$, where n is an arbitrary vector. The choice $n^2 = 0$ is called light-cone gauge. Again implemented

through Lagrange multipliers:

$$\mathcal{L}_{\mathrm{GF}} = -\frac{1}{2\lambda}\left(n_\mu \mathcal{A}^\mu\right)^2 \tag{16}$$

which leads to the propagator:

$$\Delta_{\mu\nu}(p,n) = \frac{-i}{p^2+i\epsilon}\left[g_{\mu\nu} - \frac{p_\mu n_\nu + p_\nu n_\mu}{p\cdot n} + \frac{n^2 p_\mu p_\nu}{(p\cdot n)^2}\right] . \tag{17}$$

Here, the final matrix elements are again independent of n. In this case, no ghost particles are interacting with the gauge field, and thus they can be integrated out from the Lagrangian and do not appear in the final expressions.

Extra: The Faddeev-Popov Procedure and Ghosts To integrate over physical configurations only, we insert a gauge-fixing condition $G^a(A) = 0$ (e.g., covariant gauge $G^a(A) = \partial^\mu A^a_\mu$) using the identity resolution:

$$1 = \Delta_{\mathrm{FP}}[A] \int \mathcal{D}\alpha\, \delta\left(G^a(A^\alpha)\right), \tag{18}$$

where $\Delta_{\mathrm{FP}}[A] = \det\left(\frac{\delta G^a(A^\alpha)}{\delta \alpha^b}\right)$ is the Faddeev-Popov determinant.

Factoring out the infinite volume of the gauge group $\int \mathcal{D}\alpha$, the effective path integral becomes:

$$Z = \int \mathcal{D}A\, \Delta_{\mathrm{FP}}[A]\, \delta\left(G^a(A)\right) \exp\left(i\int \mathrm{d}^4x\, \mathcal{L}_{\mathrm{inv}}[A]\right). \tag{19}$$

By shifting the condition to $G^a(A) = \omega^a(x)$ and performing a Gaussian functional integration over $\omega^a(x)$ with width parameter ξ, we generate the standard covariant gauge-fixing term:

$$\mathcal{L}_{\mathrm{gf}} = -\frac{1}{2\xi}\left(G^a(A)\right)^2 = -\frac{1}{2\xi}(\partial^\mu A^a_\mu)(\partial^\nu A^a_\nu). \tag{20}$$

The determinant $\det\left(\frac{\delta G^a}{\delta \alpha^b}\right)$ is exponentiated into the Lagrangian by introducing anti-commuting scalar fields $c^a, \bar{c}^a$ (Faddeev-Popov ghosts and antighosts):

$$\Delta_{\mathrm{FP}}[A] = \int \mathcal{D}c\, \mathcal{D}\bar{c}\, \exp\left(i\int \mathrm{d}^4x\, \mathcal{L}_{\mathrm{ghost}}\right), \tag{21}$$

where for Yang-Mills gauge theory in covariant gauges:

$$\frac{\delta G^a(A^\alpha)}{\delta \alpha^b} = -\frac{1}{g}\partial^\mu D^{ab}_\mu = -\frac{1}{g}\left(\delta^{ab}\partial^2 - g f^{abc}\partial^\mu A^c_\mu\right). \tag{22}$$

Absorbing constant normalization factors, the ghost Lagrangian reads:

$$\mathcal{L}_{\mathrm{ghost}} = -\bar{c}^a \partial^\mu D^{ab}_\mu c^b = \bar{c}^a(-\partial^2\delta^{ab} - g f^{abc}\partial^\mu A^c_\mu)c^b. \tag{23}$$

In physical gauges $\partial A = 0$ and the ghost fields decouple.

2.3 The Brout-Englert-Higgs Mechanism

To endow intermediate vector bosons with physical masses without spoiling local gauge invariance, we introduce a complex scalar doublet Φ transforming under $SU(2)_L \times U(1)_Y$ with hypercharge $Y_\Phi = 1/2$:

$$\Phi(x) = \begin{pmatrix} \phi^+(x) \\ \phi^0(x) \end{pmatrix}. \tag{24}$$

The gauge-invariant scalar Lagrangian is given by:

$$\mathcal{L}_\Phi = (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi), \tag{25}$$

where the gauge covariant derivative acting on Φ is:

$$D_\mu \Phi = \left(\partial_\mu - ig \frac{\sigma^a}{2} W_\mu^a - ig' Y_\Phi B_\mu \right) \Phi. \tag{26}$$

The self-interaction potential takes the renormalizable form:

$$V(\Phi) = \mu^2 (\Phi^\dagger \Phi) + \lambda (\Phi^\dagger \Phi)^2, \quad \lambda > 0. \tag{27}$$

When the mass parameter is chosen such that $\mu^2 < 0$, the potential develops an unstable local maximum at $\Phi = 0$ and a continuous manifold of degenerate minima at:

$$\langle \Phi^\dagger \Phi \rangle = -\frac{\mu^2}{2\lambda} \equiv \frac{v^2}{2}, \quad v \approx 246 \text{ GeV}. \tag{28}$$

Choosing a specific vacuum direction that preserves electric charge invariance ($U(1)_{\text{EM}}$ unbroken):

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \tag{29}$$

Parametrizing the scalar perturbations around this vacuum state:

$$\Phi(x) = \frac{1}{\sqrt{2}} U(\pi^a) \begin{pmatrix} 0 \\ v + H(x) \end{pmatrix}, \quad U(\pi^a) = \exp\left(\frac{i \sigma^a \pi^a(x)}{v} \right). \tag{30}$$

In the **unitary gauge**, a local $SU(2)_L$ gauge rotation is performed to eliminate the three Goldstone fields:

$$\Phi \to U^{-1}(\pi^a) \Phi \,, \tag{31}$$

$$W_\mu \to U^{-1} W_\mu U - \frac{i}{g} U^{-1} \partial_\mu U \,. \tag{32}$$

This leaves only the physical Higgs scalar field $H(x)$:

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + H(x) \end{pmatrix}. \tag{33}$$

and explicit mass terms for the vector bosons that arise from the kinematic term. Evaluating the covariant derivative on the vacuum expectation value gives rise to quadratic gauge boson bilinear terms:

$$(D_\mu\Phi)^\dagger(D^\mu\Phi) \supset \frac{1}{2}\begin{pmatrix}0 & v\end{pmatrix}\left(\frac{g}{2}\sigma^a W^a_\mu + \frac{g'}{2}B_\mu\right)\left(\frac{g}{2}\sigma^b W^{\mu,b} + \frac{g'}{2}B^\mu\right)\begin{pmatrix}0\\ v\end{pmatrix}. \tag{34}$$

Expanding this product:

$$(D_\mu\Phi)^\dagger(D^\mu\Phi) \supset \frac{v^2}{8}\left[g^2(W^1_\mu)^2 + g^2(W^2_\mu)^2 + (-gW^3_\mu + g'B_\mu)^2\right]. \tag{35}$$

Diagonalizing this mass matrix yields the physical mass eigenstates:

$$W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \mp iW^2_\mu) \qquad \Longrightarrow\ M_W = \frac{1}{2}gv, \tag{36}$$

$$Z_\mu = \cos\theta_W W^3_\mu - \sin\theta_W B_\mu \qquad \Longrightarrow\ M_Z = \frac{1}{2}\sqrt{g^2+g'^2}v = \frac{M_W}{\cos\theta_W}, \tag{37}$$

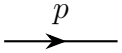
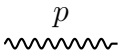
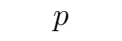
$$A_\mu = \sin\theta_W W^3_\mu + \cos\theta_W B_\mu \qquad \Longrightarrow\ M_A = 0, \tag{38}$$

where the weak mixing angle θ_W is defined by $\cos\theta_W \equiv g/\sqrt{g^2+g'^2}$.

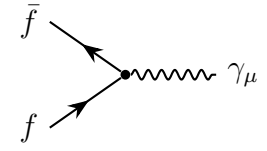
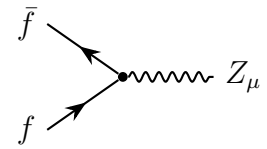
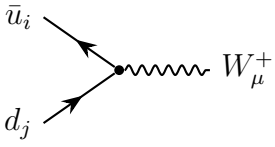
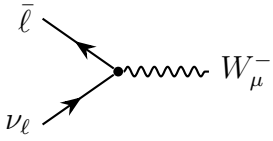
2.4 EW Feynman rules

From these expressions, we can extract the Feynman rules for the SM after EWSB. The QCD rules are in section 3. All propagators are given in the $R_\xi = 1$ gauge, i.e. Feynman gauge.

Propagators

Field / Line	Feynman Graph	Propagator Expression	
Fermion (f)		$\dfrac{i(\not{p}+m_f)}{p^2-m_f^2+i\epsilon}$	
Vector Boson ($\gamma, W^\pm, Z^0$)		$\dfrac{-ig^{\mu\nu}}{p^2-M_V^2+i\epsilon}$	$(\xi=1)$
Higgs / Goldstone (H, ϕ)		$\dfrac{i}{p^2-M^2+i\epsilon}$	

Fermion-Gauge Boson Interaction Vertices

Vertex	Feynman Graph	Feynman Rule
$f\bar{f}\gamma$		$-ieQ_f\gamma^\mu$
$f\bar{f}Z$		$-i\frac{g_W}{2\cos\theta_W}\gamma^\mu\left(v_f - a_f\gamma^5\right)$
$u_i\bar{d}_jW^+$		$-i\frac{g_W}{\sqrt{2}}V_{ij}\gamma^\mu P_L$
$\ell\bar{\nu}_\ell W^-$		$-i\frac{g_W}{\sqrt{2}}\gamma^\mu P_L$

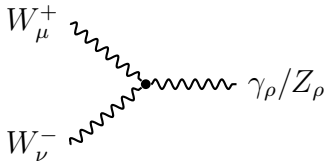
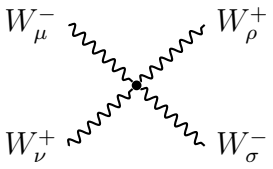
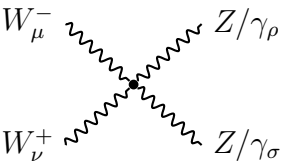
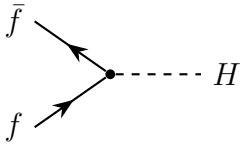
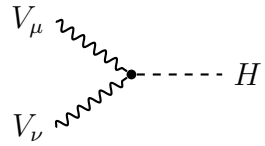
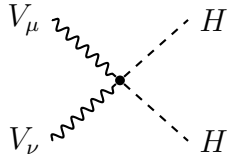
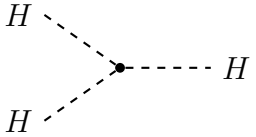
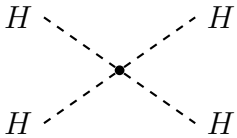
Coupling Parameters:

$$v_f = T_3^f - 2Q_f \sin^2\theta_W, \quad a_f = T_3^f$$

Triple and Quartic Gauge Boson Vertices Define the canonical kinematic 3-boson tensor (all momenta incoming):

$$V_{\mu\nu\rho}(p_1,p_2,p_3) = (p_1-p_2)_\rho g_{\mu\nu} + (p_2-p_3)_\mu g_{\nu\rho} + (p_3-p_1)_\nu g_{\rho\mu}$$

Vacuum Expectation Value (VEV): $v = \left(\sqrt{2}G_F\right)^{-1/2} = \frac{2M_W}{g_W} \approx 246$ GeV.
Coupling parameters: $g_{WW\gamma} = e$, $g_{WWZ} = g_W\cos\theta_W$, $g_{WWZZ} = g_W^2\cos^2\theta_W$, $g_{WW\gamma\gamma} = e^2$, $g_{WWZ\gamma} = eg_W\cos\theta_W$.

Vertex	Feynman Graph	Feynman Rule
$W^+W^-\gamma/Z$		$-ig_{WWV}V_{\mu\nu\rho}(p_1,p_2,p_3)$
$W^+W^-W^+W^-$		$ig_W^2(2g_{\mu\rho}g_{\nu\sigma}-g_{\mu\nu}g_{\rho\sigma}-g_{\mu\sigma}g_{\nu\rho})$
W^+W^-VV'		$-ig_{WWVV'}(2g_{\mu\nu}g_{\rho\sigma}-g_{\mu\rho}g_{\nu\sigma}-g_{\mu\sigma}g_{\nu\rho})$
Yukawa ($\bar{f}fH$)		$-i\frac{m_f}{v}=-i\frac{g_W m_f}{2M_W}$
HVV $(V=W,Z)$		$i\frac{2M_V^2}{v}g_{\mu\nu}$
$HHVV$		$i\frac{2M_V^2}{v^2}g_{\mu\nu}$
Higgs Self (H^3)		$-i\frac{3M_H^2}{v}$
Higgs Self (H^4)		$-i\frac{3M_H^2}{v^2}$

External Line Factors and Loop Rules

External Line	Factor	External Line	Factor
Incoming Fermion (s)	$u(p,s)$	Outgoing Fermion (s)	$\bar{u}(p,s)$
Incoming Anti-Fermion (s)	$\bar{v}(p,s)$	Outgoing Anti-Fermion (s)	$v(p,s)$
Incoming Vector Boson (λ)	$\varepsilon_\mu(p,\lambda)$	Outgoing Vector Boson (λ)	$\varepsilon_\mu^*(p,\lambda)$

- **Loop Integrals:** Integrate over every undetermined loop four-momentum k:

$$\int \frac{d^4k}{(2\pi)^4}$$

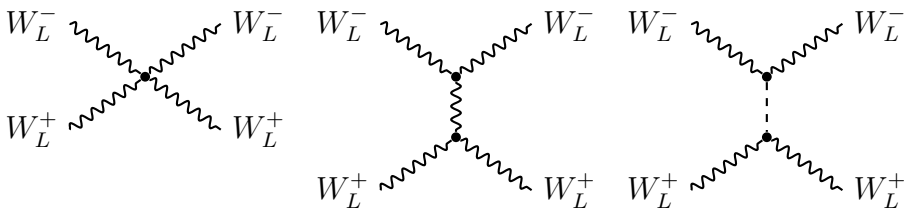


Figure 2: Feynman diagrams contributing to WW scattering.

- **Fermion Loops:** Multiply by a factor of (-1) for each closed fermion loop.
- **Identical Fermions:** Multiply by a factor of (-1) for permutations/crossing of identical external fermions.
- **Symmetry Factors:** Divide by the combinatorial symmetry factor S for identical internal field configurations.

2.5 Unitarity in Longitudinal Vector Boson Scattering

A massive spin-1 vector boson carrying 4-momentum $p^\mu = (E, 0, 0, p)$ admits three polarization states. The longitudinal polarization four-vector is:

$$\epsilon_L^\mu(p) = \left(\frac{p}{M}, 0, 0, \frac{E}{M}\right) = \frac{p^\mu}{M} + \mathcal{O}\left(\frac{M}{E}\right). \tag{39}$$

Notice that as $E \gg M$, the longitudinal polarization vector scales linearly with momentum: $\epsilon_L^\mu(p) \propto E/M$.

In the high-energy scattering of longitudinal W bosons, $W_L^+W_L^- \to W_L^+W_L^-$, tree-level amplitudes from gauge-boson exchange diagrams (γ, Z exchange and quartic contact interaction, see figure 2) grow with the center-of-mass energy squared s:

$$\mathcal{M}_{\text{gauge}} = \frac{g^2}{4M_W^2}(s+t) + \mathcal{O}(1) \propto \frac{s}{v^2}. \tag{40}$$

This violates the Froissart-Martin unitarity bound ($\lim_{s\to\infty} \sigma^{tot} \leq C \log^2(s/s_0)$) on the growth of total cross sections. However, the s- and t-channel exchange of the physical Higgs boson yields:

$$\mathcal{M}_{\text{Higgs}} = -\frac{g^2}{4M_W^2}(s+t) + \mathcal{O}(1). \tag{41}$$

The terms proportional to s and t cancel identically, leaving:

$$\mathcal{M}_{\text{total}} = \mathcal{M}_{\text{gauge}} + \mathcal{M}_{\text{Higgs}} \xrightarrow{s \gg M_W^2} -\frac{m_H^2}{v^2}\left[\frac{s}{s-m_H^2} + \frac{t}{t-m_H^2}\right] \sim \mathcal{O}(1). \tag{42}$$

Multi-boson production processes (VV, VVV, Vector Boson Scattering) directly probe this delicate gauge-Higgs cancellation mechanism. This has been studied extensively at

LEP and the LHC. Before the Higgs discovery, the interpretation was different, as large deviations from this structure were realistic. The discovery of the Higgs, of course, changed that picture completely and turned the exercise of looking into multi-boson interactions into a precision game.

To study such processes at hadron colliders, we need to go beyond the EW theory and dive into QCD, i.e., the theory of strong interactions, because it plays a prominent role in any process we might be interested in.

3 QCD Framework

3.1 Invitation: event simulations at hadron colliders

The SM is a *complicated* theory model. We can't produce exact predictions in general (except for some quite general symmetry-based predictions), especially not for high-energy scattering experiments with high-multiplicity final states. In the limit of small coupling, perturbative methods allow us to make systematic approximations of transition amplitudes and, by that, of cross sections. This underpins the success of QFT, for example in QED. QCD, the theory of the strong force, is however strongly coupled at low energies, and perturbative methods apply only at very high energies, and non-perturbative as well as all-order effects are important. At low energy, lattice QCD can provide non-perturbative results by discretizing path integrals. Although very powerful, it is currently infeasible to use at high energy, because the number of lattice points is inversely proportional to the energy scale.

In practice, we need approximations that introduce theoretical uncertainties. In the best case, we can systematically improve the approximation and thus the precision/accuracy of the predictions. This is the case for perturbative methods, for example fixed-order perturbation theory, resummation, and parton showering. In some cases this is not the case (often the non-perturbative parts of calculations), and we must hope that their impact is small for the physics of interest or can at least be systematically assessed.

Modern collider physics relies fundamentally on General-Purpose Monte Carlo event generators (notably Pythia 8, Herwig 7, and Sherpa 3). They serve as the computational bridge connecting (Fig. 3):

1. **The Theory World:** The Standard Model Lagrangian $\mathcal{L}_{\text{SM}}$, gauge symmetries, and perturbative Feynman amplitude calculations.

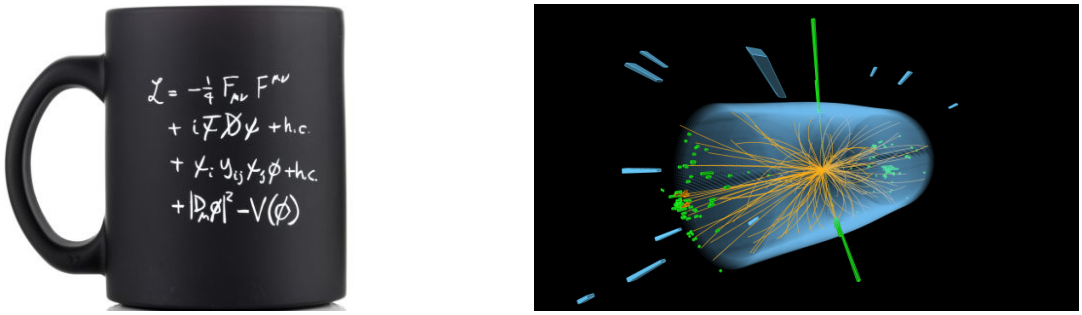


Figure 3: Theory vs Real World

2. **The Real World:** Hundreds of collimated, detectable color-singlet hadrons, leptons, and photons measured inside detectors such as ATLAS and CMS.

Monte Carlo event generators combine systematically improvable parts (fixed-order, resummation/parton-shower) with phenomenological models for non-perturbative physics (Fig. 3.1). This combination exploits scale factorization to separate the important degrees of freedom at different stages of the simulations:

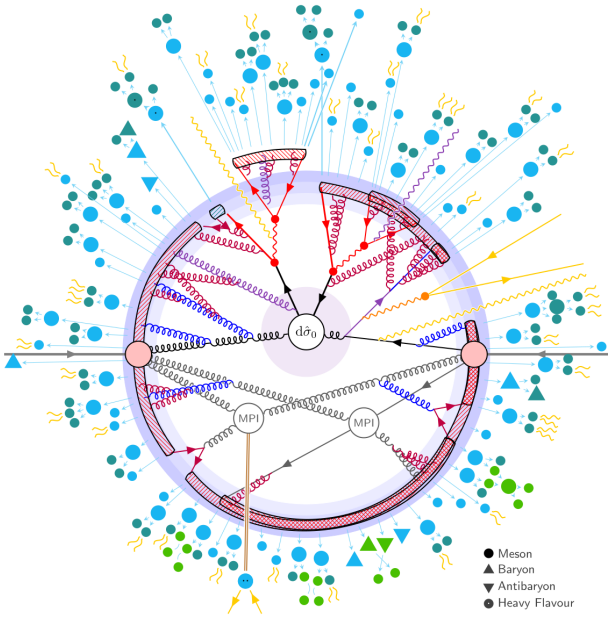
"What you see depends on the resolution scale."

Due to the asymptotic freedom of Quantum Chromodynamics (QCD), the strong coupling constant $\alpha_s(Q^2)$ runs as:

$$\alpha_s(\mu^2) = \frac{\alpha_s(\mu_0^2)}{1 + b_0 \alpha_s(\mu_0^2) \ln(\mu^2/\mu_0^2)}, \qquad b_0 = \frac{11C_A - 2n_f}{12\pi}. \tag{43}$$

This allows a systematic decomposition into four primary physical domains across decreasing energy scales (Q):

- **Hard Scattering ($Q \gg \Lambda_{\text{QCD}}$):** High momentum transfer, evaluated with fixed-order perturbation theory. Here the coupling is small, $\alpha_s(m_Z) \approx 0.118$.
- **Parton Shower ($1\ \text{GeV} \lesssim Q \lesssim \mu_F$):** Multiple soft and collinear branchings, evaluated with all-orders logarithmic resummation. This covers the regime down to scales where the strong coupling becomes non-perturbative.
- **Matching/Merging:** Eliminating double counting and phase-space gaps between fixed-order and resummed/showered predictions.
- **Hadronization & Beam Remnants ($Q \sim \Lambda_{\text{QCD}}$):** Non-perturbative confinement models (color-reconnection, fragmentation) and multi-parton interactions.



In this lecture, we focus on producing EW bosons that are generally quite heavy (except the photon) or produced with large momenta (here photons can also appear); in other words, we are interested in high energies. This means we will be mainly interested in the perturbative regime of the simulations, which is also the reason we can achieve any sense of 'precision' or even more basically a 'test of the SM' to begin with. Non-perturbative effects, except the PDFs, can matter at high precision or in corners of phase space, but we will set them aside for now.

3.2 Basics of QCD

This section introduces QCD and provides the starting point for perturbative calculations, as well as some more advanced concepts later on. During the lecture, we will skim over many details, and the notes will serve more as a reference for discussion.

3.2.1 Experimental Evidence for the Color Degree of Freedom

Quantum Chromodynamics (QCD) is the non-Abelian $SU(3)$ gauge theory governing the strong interaction among quarks and gluons. The introduction of the internal degree of freedom termed *color* ($N_c = 3$) is mandated by several critical experimental and theoretical observations:

1. **Spin-Statistics Paradox of the Δ^{++} Baryon:**
   The Δ^{++} baryon has spin $J = 3/2$ and electric charge $+2$, comprising three up quarks (uuu). In its ground state ($L = 0$), its spatial wavefunction ψ_{space}, spin

state $\chi_{\text{spin}} = |\uparrow\uparrow\uparrow\rangle$, and flavor state $\phi_{\text{flavor}} = |uuu\rangle$ are all symmetric under particle exchange:

$$|\Delta^{++}\rangle_{\text{total}} = \psi_{\text{space}} \otimes \chi_{\text{spin}} \otimes \phi_{\text{flavor}} \otimes \xi_{\text{color}}. \tag{44}$$

Since quarks are Dirac fermions, Fermi-Dirac statistics require the overall state to be totally antisymmetric. Thus, an exact $SU(3)$ color singlet representation must exist:

$$\xi_{\text{color}} = \frac{1}{\sqrt{6}}\epsilon^{ijk}|q_i q_j q_k\rangle \quad (i,j,k \in \{1,2,3\}). \tag{45}$$

2. **The R-ratio in e^+e^- Annihilation:**
   At leading order (LO) in the parton model, the ratio of total hadronic cross-section to the muon pair cross-section in e^+e^- collisions measures the sum over all active quark color-spin states:

$$R \equiv \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)} = N_c \sum_f Q_f^2, \tag{46}$$

   where Q_f is the electric charge of quark flavor f in units of the positron charge e.

   - Below the charm threshold (u, d, s active):

$$R = 3 \times \left[\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2\right] = 3 \times \frac{6}{9} = 2. \tag{47}$$

   - Above the bottom threshold (u, d, s, c, b active):

$$R = 3 \times \left(\frac{4}{9} + \frac{1}{9} + \frac{1}{9} + \frac{4}{9} + \frac{1}{9}\right) = \frac{11}{3} \approx 3.67. \tag{48}$$

3. **Neutral Pion Decay ($\pi^0 \to \gamma\gamma$):**
   From the axial anomaly (ABJ anomaly) in QED, the decay width is given by:

$$\Gamma(\pi^0 \to \gamma\gamma) = \left(\frac{N_c}{3}\right)^2 \frac{\alpha_{\text{em}}^2 m_\pi^3}{64\pi^3 f_\pi^2} \approx 7.73 \text{ eV} \quad (\text{for } N_c = 3), \tag{49}$$

   matching experimental measurements with sub-percent precision.

3.2.2 The Non-Abelian SU(3) Gauge Theory & QCD Lagrangian

Quark fields transform under the fundamental representation (**3**) of the local gauge group $SU(3)_c$:

$$\psi(x) \to \psi'(x) = U(x)\psi(x) = \exp\left(ig_s\theta^a(x)T^a\right)\psi(x), \tag{50}$$

where T^a ($a = 1, \ldots, 8$) are the 8 generators of the Lie algebra $\mathfrak{su}(3)$ satisfying:

$$[T^a, T^b] = if^{abc}T^c. \tag{51}$$

The gauge-covariant derivative acting on quarks is:

$$D_\mu = \partial_\mu - ig_s T^a A^a_\mu, \tag{52}$$

where A^a_μ are the 8 non-Abelian gluon gauge fields transforming in the adjoint representation (**8**).

The non-Abelian field strength tensor $F^a_{\mu\nu}$ is defined through the commutator of covariant derivatives:

$$[D_\mu, D_\nu] = -ig_s T^a F^a_{\mu\nu} \implies F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g_s f^{abc} A^b_\mu A^c_\nu. \tag{53}$$

The classical, gauge-invariant QCD Lagrangian density is:

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu} + \sum_{f=1}^{n_f} \bar{\psi}_{f,i} (i \not{D}_{ij} - m_f \delta_{ij}) \psi_{f,j}. \tag{54}$$

The quadratic term in $F^a_{\mu\nu}$ gives rise to cubic ($g_s f^{abc}$) and quartic ($g_s^2 f^{abe} f^{cde}$) gluon self-interactions, leading directly to the phenomenon of asymptotic freedom.

3.2.3 Color Algebra, Representation Invariants, and Fierz Identity

The standard representation uses Gell-Mann matrices: $T^a = \frac{1}{2}\lambda^a$:

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda^2 = \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \quad \lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

The core group-theoretical constants are:

- **Fundamental Index (T_F):**

$$\text{Tr}(T^a T^b) = T_F \delta^{ab}, \quad T_F = \frac{1}{2}. \tag{55}$$

- **Fundamental Casimir (C_F):**

$$(T^a T^a)_{ij} = C_F \delta_{ij}, \quad C_F = \frac{N_c^2 - 1}{2N_c} = \frac{3^2 - 1}{2 \times 3} = \frac{4}{3}. \tag{56}$$

- **Adjoint Casimir (C_A):**

$$f^{acd} f^{bcd} = C_A \delta^{ab}, \quad C_A = N_c = 3. \tag{57}$$

- **Fierz Completeness Identity:**

$$T^a_{ij}T^a_{kl} = \frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{N_c}\delta_{ij}\delta_{kl}\right). \tag{58}$$

- **Color Trace Representation of Structure Constants:**

$$\mathrm{Tr}(T^aT^bT^c) - \mathrm{Tr}(T^cT^bT^a) = iT_F f^{abc} = \frac{i}{2}f^{abc} \tag{59}$$

3.2.4 Gauge Fixing and Faddeev-Popov Ghosts

As we saw in the section on the EW theory, we need gauge fixing to derive gauge-boson propagators. In covariant R_ξ gauges:

$$\mathcal{L}_{\mathrm{gf}} = -\frac{1}{2\xi}(\partial^\mu A^a_\mu)^2. \tag{60}$$

The corresponding Faddeev-Popov determinant introduces anticommuting scalar fields (ghosts $c^a, \bar{c}^a$) transforming in the adjoint representation:

$$\mathcal{L}_{\mathrm{ghost}} = \bar{c}^a(-\partial^\mu D^{ab}_\mu)c^b = \bar{c}^a\left(-\partial^2\delta^{ab} - g_s f^{abc}\partial^\mu A^c_\mu\right)c^b. \tag{61}$$

Ghosts cancel the contributions of unphysical timelike and longitudinal gluon polarizations in loop amplitudes.

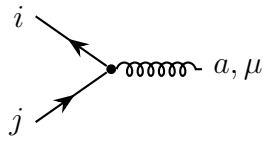
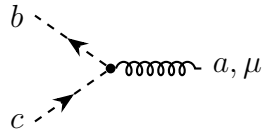
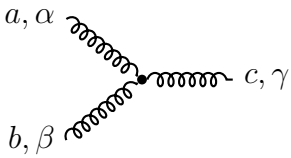
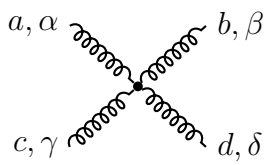
3.2.5 QCD Feynman Rules

All momenta are defined flowing **into** the vertices unless explicitly indicated otherwise. Fundamental color indices are denoted by $i, j \in \{1, \ldots, N_c\}$, and adjoint gluon color indices by $a, b, c, d \in \{1, \ldots, N_c^2 - 1\}$.

Propagators

Field / Line	Feynman Graph	Propagator Expression
Quark (q)	$j \xrightarrow{\quad p \quad} i$	$\delta_{ij}\dfrac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}$
Gluon (A^a_μ) *Cov.* R_ξ	a, μ ~~~ b, ν (p)	$\dfrac{-i\delta^{ab}}{p^2 + i\epsilon}\left[g^{\mu\nu} - (1-\xi)\dfrac{p^\mu p^\nu}{p^2}\right]$ ($\xi = 1$ for Feynman Gauge)
Gluon (A^a_μ) *Axial* ($n \cdot A = 0$)	a, μ ~~~ b, ν (p)	$\dfrac{-i\delta^{ab}}{p^2 + i\epsilon}\left[g^{\mu\nu} - \dfrac{n^\mu p^\nu + n^\nu p^\mu}{n \cdot p} + \dfrac{n^2 p^\mu p^\nu}{(n \cdot p)^2}\right]$
Ghost (c^a)	b - - - a (p)	$\delta^{ab}\dfrac{i}{p^2 + i\epsilon}$

Vertices

Vertex	Feynman Graph	Feynman Rule
Quark-Gluon		$-ig_s(T^a)_{ij}\gamma^\mu$
Ghost-Gluon		$-g_s f^{abc} p^\mu$ (p is the outgoing ghost momentum)
3-Gluon Vertex		$-g_s f^{abc} V_{\alpha\beta\gamma}(p,q,r)$
4-Gluon Vertex		$-ig_s^2\Big[f^{xac}f^{xbd}(g_{\alpha\beta}g_{\gamma\delta}-g_{\alpha\delta}g_{\beta\gamma})$ $+f^{xad}f^{xcb}(g_{\alpha\gamma}g_{\beta\delta}-g_{\alpha\beta}g_{\gamma\delta})$ $+f^{xab}f^{xdc}(g_{\alpha\delta}g_{\beta\gamma}-g_{\alpha\gamma}g_{\beta\delta})\Big]$

3.2.6 Ultraviolet Renormalization and the beta-function

Loop corrections in dimensional regularization ($D = 4 - 2\epsilon$) yield ultraviolet poles $1/\epsilon_{\text{UV}}$. In the $\overline{\text{MS}}$ scheme, the running strong coupling $\alpha_s(\mu_R^2) = \frac{g_s^2(\mu_R^2)}{4\pi}$ satisfies the Callan-Symanzik renormalization group equation:

$$\mu_R^2 \frac{\partial \alpha_s}{\partial \mu_R^2} = \beta(\alpha_s) = -\alpha_s^2 \left(b_0 + b_1\alpha_s + b_2\alpha_s^2 + \dots\right), \tag{62}$$

with the 1-loop and 2-loop coefficients:

$$b_0 = \frac{11C_A - 2n_f}{12\pi} = \frac{33 - 2n_f}{12\pi}, \tag{63}$$

$$b_1 = \frac{17C_A^2 - 5C_A n_f - 3C_F n_f}{24\pi^2} = \frac{153 - 19n_f}{24\pi^2}. \tag{64}$$

3.2.7 Asymptotic Freedom and the QCD Scale

For the physical Standard Model ($n_f \leq 6$), $b_0 > 0$. Solving the 1-loop RGE gives:

$$\alpha_s(Q^2) = \frac{\alpha_s(\mu_0^2)}{1 + b_0\alpha_s(\mu_0^2)\ln\left(\frac{Q^2}{\mu_0^2}\right)} \equiv \frac{1}{b_0 \ln\left(\frac{Q^2}{\Lambda_{\text{QCD}}^2}\right)}, \tag{65}$$

where $\Lambda_{\text{QCD}} \approx 200$–$300$ MeV is the non-perturbative scale parameter where the perturbative coupling diverges (Landau pole).

- **Asymptotic Freedom ($Q^2 \gg \Lambda_{\text{QCD}}^2$):** $\alpha_s(Q^2) \to 0$. Quarks and gluons act as quasi-free particles; high-energy cross-sections are perturbatively calculable.

- **Confinement ($Q^2 \sim \Lambda^2_{\mathbf{QCD}}$):** $\alpha_s(Q^2) \sim \mathcal{O}(1)$. Non-perturbative dynamics dominate; isolated color charges cannot propagate as free asymptotic states.

World Average Reference Value: $\alpha_s(M_Z) = 0.1180 \pm 0.0009$.

3.3 Hadron Structure, Deep Inelastic Scattering, and PDFs

3.3.1 Deep Inelastic Scattering

Deep Inelastic Scattering (DIS) of a charged lepton off a hadron (e.g., $e^- + P \to e^- + X$) provides the cleanest probe of hadronic substructure and the field-theoretic definition of Parton Distribution Functions (PDFs).

Let k and k' denote the incoming and scattered lepton four-momenta, and let P denote the target proton four-momentum ($P^2 = M^2$).

Kinematic Invariants

- **Momentum Transfer / Virtuality (Hard Resolving Scale):**

$$q^\mu = k^\mu - k'^\mu, \qquad Q^2 \equiv -q^2 = -(k - k')^2 > 0, \tag{66}$$

  resolving transverse spatial distances inside the hadron $\Delta r_\perp \sim 1/Q$.

- **Bjorken Scaling Variable:**

$$x \equiv \frac{Q^2}{2P \cdot q} \in [0, 1]. \tag{67}$$

  In the infinite-momentum frame, x represents the fraction of the hadron's longitudinal momentum carried by the struck parton.

- **Inelasticity:**

$$y \equiv \frac{P \cdot q}{P \cdot k} \in [0, 1], \tag{68}$$

  describing the lepton's fractional energy loss in the target proton rest frame.

- **Center-of-Mass Energy Relation:**

$$s \approx 2P \cdot k \implies Q^2 = xys. \tag{69}$$

Leptonic and Hadronic Tensors The unpolarized differential cross section factorizes into a calculable QED leptonic tensor $L^{\mu\nu}$ and a non-perturbative hadronic tensor $W_{\mu\nu}$:

$$\mathrm{d}\sigma = \sum_X \frac{1}{4ME} \int \mathrm{d}\Phi \frac{1}{4} \sum_{\text{pol}} |\mathcal{M}|^2 = \frac{1}{4ME} \frac{\mathrm{d}^3k'}{(2\pi)^3 2E_{k'}} \frac{e^4}{Q^4} L^{\mu\nu} W_{\mu\nu}. \tag{70}$$

The leptonic tensor for an unpolarized Dirac fermion (neglecting lepton mass) is:

$$L_{\mu\nu} = 2\left(k_\mu k'_\nu + k'_\mu k_\nu - g_{\mu\nu}(k \cdot k')\right). \tag{71}$$

Using the decomposition of the phase space $\mathrm{d}\Phi = \frac{\mathrm{d}^3\vec{k}'}{2E'(2\pi)^2}\mathrm{d}\Phi_X$ as well as of the matrix element $\frac{1}{4}\sum_{\text{pol}} |\mathcal{M}|^2 = \frac{e^4}{Q^4} L^{\mu\nu} H_{\mu\nu}$. The hadronic tensor is then defined as the inclusive sum over hadronic final states:

$$W_{\mu\nu} = \frac{1}{8\pi} \sum_X \int \mathrm{d}\Phi_X H_{\mu\nu}. \tag{72}$$

Electromagnetic gauge invariance ($q^\mu W_{\mu\nu} = 0$) and parity conservation (for photon exchange only) restrict the parameterization of $W_{\mu\nu}$ to two independent structure functions $W_1(x, Q^2)$ and $W_2(x, Q^2)$:

$$W_{\mu\nu}(P,q) = \left(-g_{\mu\nu} + \frac{q_\mu q_\nu}{q^2}\right) W_1(x,Q^2) + \left(P_\mu - q_\mu \frac{P\cdot q}{q^2}\right)\left(P_\nu - q_\nu \frac{P\cdot q}{q^2}\right) \frac{W_2(x,Q^2)}{P\cdot q}. \tag{73}$$

3.3.2 The Quark-Parton Model and Bjorken Scaling

In the Bjorken limit ($Q^2, P\cdot q \to \infty$ with x fixed), internal non-perturbative bindings operate on timescales much longer than the interaction time $\tau \sim 1/Q$. Partons act as free, non-interacting Dirac fermions.

Structure Functions and the Callan-Gross Relation After contraction with the leptonic tensore we have

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{4\pi\alpha}{yQ^4}\left[y^2 W_1(x,Q^2) + \left(\frac{1-y}{x} - xy\frac{M^2}{Q^2}\right) W_2(x,Q^2)\right]. \tag{74}$$

In the limit $Q^2 \to \infty$ for fixed x, and reorganisation (we relabel W into F for this limit) of the terms we have

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{4\pi\alpha}{yQ^4}\left[(1+(1-y)^2)F_1 + \frac{1-y}{x}(F_2 - 2xF_1)\right] \tag{75}$$

Because quarks carry spin-1/2 and scatter via transverse photons, the longitudinal structure function vanishes in the collinear limit:

$$F_L(x) \equiv F_2(x) - 2xF_1(x) = 0 \implies F_2(x) = 2xF_1(x) \quad (\textbf{Callan-Gross Relation}). \tag{76}$$

This can also directly derived in the parton model. In reality however we see that we have a Q^2 dependence, in particular at small x (scaling violations).

Parton Interpretation We can redo the calculus above for the case of a quark scattering with an electron. Let assume that the quarks momentum is $p = \omega P$, where P is the momentum we had for the proton and $\omega \in [0, 1]$. We can calculate now the matrix element

$$\frac{1}{4}\sum_{\text{pol}} |\mathcal{M}|^2 = \frac{e^4 e_f^2}{Q^4} L^{\mu\nu} Q_{\mu\nu} \ , \quad Q^{\mu\nu} = p^\mu p'^\nu + p^\nu p'^\mu - g^{\mu\nu}(p \cdot p') \tag{77}$$

and parameterize the phase space explicitly

$$\mathrm{d}\Phi_2 = \frac{\mathrm{d}^3 k'}{(2\pi)^3 2E'} \frac{\mathrm{d}^4 p'}{(2\pi)^3} \delta(p'^2)(2\pi)^4 \delta^4(k + p - k' - p') = \frac{\mathrm{d}\phi}{(4\pi)^2} \mathrm{d}x \mathrm{d}y \delta(\omega - x) \ . \tag{78}$$

We find then for the cross section

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{4\pi\alpha}{yQ^4} \left[1 + (1-y)^2\right] \frac{1}{2} e_f^2 \delta(\omega - x) \ . \tag{79}$$

Here we see that the longitudinal term proportional to $(1 - y)/x$ does not even appear (we showed effectively the Callan Gross relation for quarks).

Naive quark model If, at leading order, we interpret the $f_i(\omega)$ as the probability to find a quark of flavor i with momentum fraction ω, then we can write the hadronic object F_2.

$$F_2(x) = \sum_i \int \mathrm{d}\omega f_i(\omega) x e_{f_i}^2 \delta(x - \omega) = \sum_i e_{f_i}^2 f_i(x) x \tag{80}$$

The $f_i(x)$ are called the Parton Distribution Function (PDF), representing the probability density of finding a parton of flavor i carrying momentum fraction x. Proton quantum numbers impose valence sum rules:

$$\int_0^1 \mathrm{d}x \left[u(x) - \bar{u}(x)\right] = 2, \tag{81}$$

$$\int_0^1 \mathrm{d}x \left[d(x) - \bar{d}(x)\right] = 1, \tag{82}$$

$$\int_0^1 \mathrm{d}x \left[s(x) - \bar{s}(x)\right] = 0. \tag{83}$$

3.3.3 QCD Corrections and Collinear Factorization

Radiative QCD emissions break exact Bjorken scaling, yielding logarithmic scale violations: $F_2(x) \to F_2(x, \mu_F^2)$.

Collinear Factorization Formula

$$F_2(x,Q^2) = \sum_{j=q,\bar{q},g} \int_x^1 \frac{\mathrm{d}z}{z} C_{2,j}\left(\frac{x}{z}, \frac{Q^2}{\mu_F^2}, \alpha_s(\mu_R^2)\right) f_j(z,\mu_F^2) + \mathcal{O}\left(\frac{\Lambda^2_{\mathrm{QCD}}}{Q^2}\right). \tag{84}$$

The DGLAP Evolution Equations The scale evolution of PDFs is governed by the coupled Integro-Differential Dokshitzer-Gribov-Lipatov-Altarelli-Parisi (DGLAP) equations:

$$\frac{\partial}{\partial \ln \mu_F^2}\begin{pmatrix} f_{q_i}(x,\mu_F^2) \\ f_g(x,\mu_F^2)\end{pmatrix} = \frac{\alpha_s(\mu_F^2)}{2\pi}\int_x^1 \frac{\mathrm{d}z}{z}\begin{pmatrix} P_{qq}\left(\frac{x}{z}\right) & P_{qg}\left(\frac{x}{z}\right) \\ P_{gq}\left(\frac{x}{z}\right) & P_{gg}\left(\frac{x}{z}\right)\end{pmatrix}\begin{pmatrix} f_{q_i}(z,\mu_F^2) \\ f_g(z,\mu_F^2)\end{pmatrix}. \tag{85}$$

Splitting Kernels and the Plus-Prescription The leading-order (LO) Altarelli-Parisi splitting kernels are:

$$P_{qq}(z) = C_F\left[\frac{1+z^2}{(1-z)_+} + \frac{3}{2}\delta(1-z)\right], \tag{86}$$

$$P_{qg}(z) = T_F\left[z^2 + (1-z)^2\right], \tag{87}$$

$$P_{gq}(z) = C_F\left[\frac{1+(1-z)^2}{z}\right], \tag{88}$$

$$P_{gg}(z) = 2C_A\left[\frac{z}{(1-z)_+} + \frac{1-z}{z} + z(1-z)\right] + \left(\frac{11C_A - 4n_f T_F}{6}\right)\delta(1-z). \tag{89}$$

Soft gluon divergences at $z \to 1$ are regularized via the plus-distribution:

$$\int_0^1 \mathrm{d}z \frac{g(z)}{(1-z)_+} \equiv \int_0^1 \mathrm{d}z \frac{g(z)-g(1)}{1-z}. \tag{90}$$

PDF fits PDFs have to be determined from data. Many aspects:

- Perturbative order
- Higher twist
- Resummation
- Heavy mass flavour scheme
- Parametrisation
- Fitting scale
- Treatment of missing higher order uncertainties
- Electroweak parameters + corrections

- Quark masses
- Evolution settings
- Fitting technique

3.4 Collinear Factorization in hadron-hadron collisions

At hadron colliders such as the LHC, cross sections for the production of a final state Y (we will be interested in $Y = VV$ boson pair production or $Y = VVjj$ VBS/VBF) factorize into universal non-perturbative Parton Distribution Functions (PDFs) and infrared-safe partonic hard-scattering cross sections:

$$\mathrm{d}\sigma_{pp\to Y+X} = \sum_{a,b} \int_0^1 \mathrm{d}x_1 \mathrm{d}x_2 \, f_a(x_1, \mu_F^2) \, f_b(x_2, \mu_F^2) \, \mathrm{d}\hat{\sigma}_{ab}(x_1, x_2, \mu_R^2, \mu_F^2), \tag{91}$$

where μ_F and μ_R denote the factorization and renormalization scales. The partonic cross section is perturbatively expanded in powers of the strong coupling $\alpha_s(\mu_R)$:

$$\mathrm{d}\hat{\sigma}_{ab} = \mathrm{d}\hat{\sigma}_{ab}^{(0)} + \underbrace{\mathrm{d}\hat{\sigma}_{ab}^{(1)}}_{\mathcal{O}(\alpha_s)} + \underbrace{\mathrm{d}\hat{\sigma}_{ab}^{(2)}}_{\mathcal{O}(\alpha_s^2)} + \dots \tag{92}$$

The leading-order cross section is described by the Born matrix element

$$d\hat{\sigma}_{ab}^{(0)} = d\hat{\sigma}_{ab}^{\mathrm{B}} = \frac{1}{2\hat{s}} \frac{1}{N_{ab}} \mathrm{d}\Phi_Y \, \langle \mathcal{M}_{ab\to Y}^{(0)} | \mathcal{M}_{ab\to Y}^{(0)} \rangle (\Phi_Y) \mathcal{F}(\Phi_Y) \, . \tag{93}$$

Here we denote $\hat{s} = x_1 x_2 s^{\mathrm{had}}$ as the partonic center of mass energy. We also introduce our notation for squared matrix elements $\langle \mathcal{M}_{ab\to Y}^{(0)} | \mathcal{M}_{ab\to Y}^{(0)} \rangle$ which imply sums over all external polarizations and colors as well as the "measurement function" $\mathcal{F}(\Phi_Y)$ which describes the observable of interest and can be used to implement phase space cutoffs and/or histograms.

The scattering amplitudes $|\mathcal{M}_{ab\to Y}\rangle$ can be calculated from perturbation theory and also obey a perturbative expansion:

$$|\mathcal{M}_{ab\to Y}\rangle = |\mathcal{M}_{ab\to Y}^{(0)}\rangle + |\mathcal{M}_{ab\to Y}^{(1)}\rangle + |\mathcal{M}_{ab\to Y}^{(2)}\rangle + \dots \tag{94}$$

3.5 Structure of NLO Calculations

The next-to-leading (NLO) coefficient reads:

$$d\hat{\sigma}_{ab}^{(1)} = d\hat{\sigma}_{ab}^{\mathrm{R}} + d\hat{\sigma}_{ab}^{\mathrm{V}} + d\hat{\sigma}_{ab}^{\mathrm{C}} \, , \tag{95}$$

where we have

$$d\hat{\sigma}^{\mathrm{R}}_{ab} = \frac{1}{2\hat{s}}\frac{1}{N_{ab}}\int \mathrm{d}\Phi_{Y+1}\,\langle \mathcal{M}^{(0)}_{ab\to Y+1}|\mathcal{M}^{(0)}_{ab\to Y+1}\rangle\,\mathcal{F}(\Phi_{Y+1})\,, \tag{96}$$

$$d\hat{\sigma}^{\mathrm{V}}_{ab} = \frac{1}{2\hat{s}}\frac{1}{N_{ab}}\int \mathrm{d}\Phi_{Y}\,2\,\mathrm{Re}\,\langle \mathcal{M}^{(0)}_{ab\to Y}|\mathcal{M}^{(1)}_{ab\to Y}\rangle\,\mathcal{F}(\Phi_{Y})\,,$$

$$d\hat{\sigma}^{\mathrm{C}}_{ab} = \frac{\alpha_s}{2\pi}\frac{1}{\epsilon}\left(\frac{\mu_R^2}{\mu_F^2}\right)^{\epsilon}\sum_c\int_0^1 \mathrm{d}z\left[P^{(0)}_{ca}(z)\,\hat{\sigma}^{\mathrm{B}}_{cb}(zx_1,x_2) + P^{(0)}_{cb}(z)\,\hat{\sigma}^{\mathrm{B}}_{ac}(x_1,zx_2)\right]\,.$$

Crucially, the NLO QCD cross section involves two distinct phase spaces:

$$d\sigma^{\mathrm{NLO}} \ni \int_Y \mathrm{d}\sigma^{\mathrm{V}} + \int_Y \mathrm{d}\sigma^{\mathrm{R}}. \tag{97}$$

Using dimensional regularization ($D = 4 - 2\epsilon$):

- **Virtual corrections ($\mathrm{d}\sigma^{\mathbf{V}}$):** Contain loop integrals with ultraviolet poles (renormalized) and soft/collinear infrared poles ($\sim 1/\epsilon^2, 1/\epsilon$).
- **Real emissions ($\mathrm{d}\sigma^{\mathbf{R}}$):** Involve phase-space integration of tree-level $(m+1)$-body final states with unresolved soft and collinear radiation.

By the Kinoshita-Lee-Nauenberg (KLN) theorem, final-state infrared divergences cancel when integrated over degenerate states:

$$\mathrm{d}\sigma^{\mathrm{V}} + \int_1 \mathrm{d}\sigma^{\mathrm{R}} = \text{finite}. \tag{98}$$

Remaining initial-state collinear poles are factorized into the scale-dependent PDFs via the DGLAP evolution equations:

$$f_a(x,\mu_F^2) = f_a(x) - \frac{1}{\epsilon}\frac{\alpha_s}{2\pi}\sum_b\int_x^1 \frac{\mathrm{d}z}{z}P_{ab}(z)f_b(x/z). \tag{99}$$

3.6 An NLO example: electron-positron annihilation to hadrons

To explicitly see the cancellation of IR divergences, let's have a look at the example of electron-positron annihilation to hadrons. This process at next-to-leading order is described by the two partonic processes: $e^+(p_a) + e^-(p_b) \to \gamma^*(q) \to q(p_1) + \bar{q}(p_2)$ and $e^+(p_a) + e^-(p_b) \to \gamma^*(q) \to q(p_1) + \bar{q}(p_2) + (g(p_3))$. This process is the primary testing ground for perturbative QCD but is also simple enough to study on the blackboard in detail. The cross-section through single-photon exchange (lowest order in the electromagnetic coupling) factorizes into leptonic and hadronic tensors:

$$\sigma(e^+e^-\to X) = \frac{1}{2s}\frac{e^2}{s^2}L_{\mu\nu}H^{\mu\nu}. \tag{100}$$

The center-of-mass energy squared is given by

$$s = q^2 = (p_a + p_b)^2 \,, \tag{101}$$

and where the spin-averaged leptonic tensor for massless initial-state leptons is:

$$L_{\mu\nu} = \frac{1}{4}\,\mathrm{Tr}\left[\not{p}_b \gamma_\mu \not{p}_a \gamma_\nu\right] = p_{a\mu}p_{b\nu} + p_{a\nu}p_{b\mu} - g_{\mu\nu}(p_a \cdot p_b). \tag{102}$$

Contracting with the conserved electromagnetic current ($q^\mu H_{\mu\nu} = 0$), we have $H^{\mu\nu} = (q^\mu q^\nu - s g^{\mu\nu})H(s)$, which yields:

$$L_{\mu\nu}H^{\mu\nu} = \frac{s}{3}\left(-g_{\mu\nu}H^{\mu\nu}\right). \tag{103}$$

Thus, the total cross section is proportional to the scalar trace $-g_{\mu\nu}H^{\mu\nu}$. All QCD corrections happen in $H^{\mu\nu}$ and thus we write:

$$H^{\mu\nu} = H^{(0)\,\mu\nu} + H^{(1)\,\mu\nu} + \mathcal{O}\big(\alpha_s^2\big) \tag{104}$$

where

$$H^{(0)\,\mu\nu} = \langle \mathcal{M}^{(0)\,\mu} | \mathcal{M}^{(0)\,\nu} \rangle \tag{105}$$

$$H^{(1)\,\mu\nu} = 2\,\mathrm{Re}(\langle \mathcal{M}^{(0)\,\mu} | \mathcal{M}^{(1)\,\nu} \rangle) \tag{106}$$

$$\tag{107}$$

3.6.1 Leading Order (LO) Born Cross-Section

To evaluate the total cross section in CDR, we need the two-body phase space in $d = 4-2\epsilon$ dimensions. The Lorentz-invariant n-body phase space in d dimensions is:

$$d\Phi_n(q; p_1, \ldots, p_n) = (2\pi)^d \delta^{(d)}\left(q - \sum_{i=1}^n p_i\right) \prod_{i=1}^n \frac{d^{d-1}p_i}{(2\pi)^{d-1} 2E_i}. \tag{108}$$

For $n = 2$ massless partons in the center-of-mass frame we can also calculate easily the total phase space volume:

$$\Phi_2 = \int d\Phi_2 = \frac{1}{8\pi}\left(\frac{4\pi}{s}\right)^\epsilon \frac{\Gamma(1-\epsilon)}{\Gamma(2-2\epsilon)}. \tag{109}$$

In $d = 4$ ($\epsilon = 0$), this reduces to the familiar $\Phi_2 = \frac{1}{8\pi}$. The Born amplitude for $\gamma^*(q) \to q(p_1)\bar{q}(p_2)$ is:

$$\mathcal{M}^{(0)\,\mu} = ie e_q \bar{u}(p_1)\gamma^\mu v(p_2)\,\delta_{ij}, \tag{110}$$

where $i, j = 1, \ldots, N_c$ are color indices. Summing over colors and spins:

$$-g_{\mu\nu}H_0^{\mu\nu} = N_c e^2 e_q^2\,\mathrm{Tr}\left[\not{p}_1 \gamma^\mu \not{p}_2 \gamma_\mu\right] = 4(1-\epsilon)N_c e^2 e_q^2 (2p_1 \cdot p_2) = 4(1-\epsilon)N_c e^2 e_q^2 s. \tag{111}$$

Including the flux factor $\frac{1}{2s}$ and leptonic contraction we have in $d = 4$:

$$\sigma_0 \equiv \sigma^{\mathrm{LO}}(e^+e^- \to q\bar{q}) = \frac{4\pi\alpha_{\mathrm{em}}^2}{3s} N_c \sum_q e_q^2 = \sigma(e^+e^- \to \mu^+\mu^-) \times N_c \sum_q e_q^2. \tag{112}$$

3.6.2 Virtual Corrections

At NLO, the virtual corrections arise from the interference between the 1-loop and the tree-level amplitude. Working in $d = 4 - 2\epsilon$, the one-loop amplitude $\mathcal{M}^{(1)\,\mu}$ receives corrections from three diagrams: the vertex-correction and the quark self-energies. For massless on-shell quarks ($p^2 = 0$), the 1-loop self-energy integral $\Sigma(p)$ is scaleless:

$$\Sigma(\not{p}) \propto \int \frac{d^D k}{(2\pi)^D} \frac{1}{k^2 (p-k)^2} \propto \frac{1}{\epsilon_{\text{UV}}} - \frac{1}{\epsilon_{\text{IR}}} = 0. \tag{113}$$

In dimensional regularization, scaleless integrals vanish because UV and IR divergences cancel algebraically. The wave-function renormalization constant in the on-shell scheme is $Z_2 = 1$. Consequently, the UV divergence is completely contained in the vertex correction and cancels against the bare electromagnetic current conservation ($Z_1 = Z_2$).

$$\mathcal{M}^{(1)\,\mu} = -i g_s^2 \mu^{2\epsilon} (T^a T^a)_{ij} \int \frac{d^D k}{(2\pi)^D} \frac{\bar{u}(p_1)\gamma^\alpha (\not{p}_1 - \not{k})\gamma^\mu (-\not{p}_2 - \not{k})\gamma_\alpha v(p_2)}{[k^2 + i\epsilon][(p_1 - k)^2 + i\epsilon][(p_2 + k)^2 + i\epsilon]}. \tag{114}$$

Using the color Casimir $\sum_a (T^a T^a)_{ij} = C_F \delta_{ij}$ where $C_F = \frac{N_c^2 - 1}{2N_c} = \frac{4}{3}$, the Dirac numerator algebra simplifies in D dimensions ($\gamma^\alpha \not{a} \gamma_\alpha = -2(1-\epsilon)\not{a}$):

$$\mathcal{N}^\mu = -2(1-\epsilon)\not{k}\gamma^\mu \not{k} + 4(p_1 \cdot p_2)\gamma^\mu - 4(p_1 \cdot k)\gamma^\mu + 4(p_2 \cdot k)\gamma^\mu - 4k^\mu(\not{p}_1 - \not{p}_2). \tag{115}$$

The terms with a single momentum k vanish due to the symmetry after integration over k^μ from $-\infty$ to ∞. The term with k^2 cancels against the denominator, and we receive a simple bubble integral.

Derivation: Massless Scalar Bubble Integral $B_0(p^2, 0, 0)$ We compute the dimensionally regularized scalar 2-point function in $D = 4 - 2\epsilon$ dimensions:

$$B_0(p^2, 0, 0) = \mu^{2\epsilon} \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{1}{k^2 (k+p)^2}. \tag{116}$$

1. **Step 1: Feynman Parametrization.** Using the identity $\frac{1}{AB} = \int_0^1 \mathrm{d}x \frac{1}{[xA + (1-x)B]^2}$:

$$\frac{1}{k^2 (k+p)^2} = \int_0^1 \mathrm{d}x \frac{1}{[x(k+p)^2 + (1-x)k^2]^2} = \int_0^1 \mathrm{d}x \frac{1}{[k^2 + 2x(k \cdot p) + xp^2]^2}. \tag{117}$$

2. **Step 2: Momentum Shift.** Shift the loop momentum $\ell^\mu = k^\mu + xp^\mu$. The denominator becomes:

$$\ell^2 - \Delta, \quad \text{with } \Delta = -x(1-x)p^2 - i\delta. \tag{118}$$

Thus, the integral becomes:

$$B_0(p^2, 0, 0) = \mu^{2\epsilon} \int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^D \ell}{(2\pi)^D} \frac{1}{[\ell^2 - \Delta]^2}. \tag{119}$$

3. **Step 3: Wick Rotation to Euclidean Space.** Rotating to Euclidean momentum $\ell^0 = i\ell_E^0$ such that $\ell^2 = -\ell_E^2$:

$$\int \frac{\mathrm{d}^D\ell}{(2\pi)^D}\frac{1}{[\ell^2-\Delta]^2} = i\int \frac{\mathrm{d}^D\ell_E}{(2\pi)^D}\frac{1}{[\ell_E^2+\Delta]^2} = \frac{i}{(4\pi)^{D/2}}\frac{\Gamma(2-D/2)}{\Gamma(2)}\Delta^{D/2-2}. \quad (120)$$

4. **Step 4: Laurent Expansion in $\epsilon = (4-D)/2$.** Substituting $D = 4-2\epsilon$:

$$B_0(p^2,0,0) = \frac{i}{(4\pi)^2}(4\pi\mu^2)^\epsilon\Gamma(\epsilon)\int_0^1 \mathrm{d}x\left[-x(1-x)p^2 - i\delta\right]^{-\epsilon}. \quad (121)$$

Expanding $\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon)$ and $A^{-\epsilon} = 1 - \epsilon\ln A + \mathcal{O}(\epsilon^2)$:

$$B_0(p^2,0,0) = \frac{i}{(4\pi)^2}\left[\frac{1}{\epsilon} - \gamma_E + \ln(4\pi) + \ln\left(\frac{\mu^2}{-p^2-i\delta}\right) - \int_0^1 \mathrm{d}x\ln[x(1-x)] + \mathcal{O}(\epsilon)\right]. \quad (122)$$

Since $\int_0^1 \mathrm{d}x\ln[x(1-x)] = -2$, we obtain the final result:

$$B_0(p^2,0,0) = \frac{i}{(4\pi)^2}\left[\frac{1}{\bar{\epsilon}} + \ln\left(\frac{\mu^2}{-p^2-i\delta}\right) + 2 + \mathcal{O}(\epsilon)\right], \quad (123)$$

where $\frac{1}{\bar{\epsilon}} \equiv \frac{1}{\epsilon} - \gamma_E + \ln(4\pi)$ is the standard $\overline{\mathrm{MS}}$ pole.

For the remaining integrals, we can use the standard three-propagator Feynman parameter representation:

$$\frac{1}{ABC} = 2\int_0^1 dx\int_0^{1-x} dy\frac{1}{[xA+yB+(1-x-y)C]^3}, \quad (124)$$

with the shifted loop momentum $\ell = k - xp_1 + yp_2$, the denominator becomes $[\ell^2 - \Delta + i\epsilon]^3$ where $\Delta = -xys - i\epsilon$.

Integrating over ℓ using:

$$\int \frac{d^D\ell}{(2\pi)^D}\frac{1}{[\ell^2-\Delta]^3} = \frac{-i}{(4\pi)^{D/2}}\frac{\Gamma(3-D/2)}{2\,\Gamma(3)}\Delta^{D/2-3} = \frac{-i}{2(4\pi)^2}(4\pi)^\epsilon\frac{\Gamma(1+\epsilon)}{\Delta^{1+\epsilon}}, \quad (125)$$

we obtain the unrenormalized 1-loop vertex form factor:

$$F_1(s) = -\frac{\alpha_s C_F}{2\pi}\frac{\Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}\left(\frac{4\pi\mu^2}{-s-i\epsilon}\right)^\epsilon\left[\frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + 4 + \mathcal{O}(\epsilon)\right]. \quad (126)$$

Putting everything together (the algebra is left to the reader `;-)`), the virtual contribution to the cross section is:

$$\sigma^{\mathrm{V}} = \sigma_0\frac{\alpha_s C_F}{2\pi}\left(\frac{4\pi\mu^2}{s}\right)^\epsilon\frac{\Gamma(1-\epsilon)^2\Gamma(1+\epsilon)}{\Gamma(1-2\epsilon)}\left[-\frac{2}{\epsilon^2} - \frac{3}{\epsilon} - 8 + \frac{7\pi^2}{6} + \mathcal{O}(\epsilon)\right]. \quad (127)$$

The virtual contribution contains:

- A double pole $-2/\epsilon^2$ corresponding to the soft-and-collinear region.
- A single pole $-3/\epsilon$ corresponding to the hard-collinear collinear region.

3.6.3 Real Emission Corrections

For $e^+e^- \to q(p_1) + \bar{q}(p_2) + g(p_3)$, let $x_i = \frac{2p_i \cdot q}{s}$ be the energy fractions in the CM frame ($x_1 + x_2 + x_3 = 2$). The invariant masses of pairs are:

$$s_{12} = s(1 - x_3), \quad s_{13} = s(1 - x_2), \quad s_{23} = s(1 - x_1). \tag{128}$$

The 3-body phase space in $D = 4 - 2\epsilon$ dimensions parameterized by x_1, x_2 is:

$$d\Phi_3 = \frac{s^{1-2\epsilon}}{32(2\pi)^3} \left(\frac{4\pi}{s}\right)^{2\epsilon} \frac{1}{\Gamma(2-2\epsilon)} (1-x_1)^{-\epsilon}(1-x_2)^{-\epsilon}(x_1+x_2-1)^{-\epsilon} dx_1 dx_2. \tag{129}$$

The physical Dalitz region is defined by:

$$0 \le x_1, x_2 \le 1, \quad x_1 + x_2 \ge 1. \tag{130}$$

The matrix element, summed over all polarizations and colors for $\gamma^* \to q\bar{q}g$:

$$|\mathcal{M}_R|^2 = 4(1-\epsilon) g_s^2 \mu^{2\epsilon} e^2 e_q^2 N_c C_F \left[\frac{x_1^2 + x_2^2 - \epsilon(2 - x_1 - x_2)^2}{(1-x_1)(1-x_2)}\right]. \tag{131}$$

Combining the phase space and matrix element:

$$\begin{aligned}\sigma^{\text{real}} =& \sigma_0 \frac{\alpha_s C_F}{2\pi} \left(\frac{4\pi\mu^2}{s}\right)^{\epsilon} \frac{\Gamma(1-\epsilon)}{\Gamma(2-3\epsilon)} \\ & \int_0^1 dx_1 \int_{1-x_1}^1 dx_2 \frac{x_1^2 + x_2^2 - \epsilon(2-x_1-x_2)^2}{(1-x_1)^{1+\epsilon}(1-x_2)^{1+\epsilon}(x_1+x_2-1)^{\epsilon}}.\end{aligned} \tag{132}$$

Explicit Integration of the real cross-section To evaluate the singular integral, let $x_2 = 1 - v(1-x_1)$ where $v \in [0,1]$:

$$\int_0^1 dx_1 (1-x_1)^{-1-2\epsilon} \int_0^1 dv\, v^{-1-\epsilon}(1-v)^{-\epsilon} \left[x_1^2 + (1 - v(1-x_1))^2 - \epsilon(1-x_1)^2(1+v)^2\right]. \tag{133}$$

Using the Euler Beta function $B(a,b) = \int_0^1 dt\, t^{a-1}(1-t)^{b-1} = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$:

$$\sigma^{\text{real}} = \sigma_0 \frac{\alpha_s C_F}{2\pi} \left(\frac{4\pi\mu^2}{s}\right)^{\epsilon} \frac{\Gamma(1-\epsilon)^2\Gamma(1+\epsilon)}{\Gamma(1-2\epsilon)} \left[\frac{2}{\epsilon^2} + \frac{3}{\epsilon} + \frac{19}{2} - \frac{7\pi^2}{6} + \mathcal{O}(\epsilon)\right]. \tag{134}$$

3.6.4 Total NLO Cross-Section & The KLN Cancellation

Combining the virtual and real corrections, we observe an exact, algebraic cancellation of all double ($1/\epsilon^2$) and single ($1/\epsilon$) infrared poles:

$$\begin{aligned}\sigma^{\text{NLO}} &= \sigma_0 + \sigma^{\text{virtual}} + \sigma^{\text{real}} \\ &= \sigma_0 \left\{1 + \frac{\alpha_s C_F}{2\pi}\left[\left(-\frac{2}{\epsilon^2} - \frac{3}{\epsilon} - 8 + \frac{7\pi^2}{6}\right) + \left(\frac{2}{\epsilon^2} + \frac{3}{\epsilon} + \frac{19}{2} - \frac{7\pi^2}{6}\right)\right]\right\} \\ &= \sigma_0 \left\{1 + \frac{\alpha_s C_F}{2\pi}\left(\frac{19}{2} - 8\right)\right\} \\ &= \sigma_0 \left(1 + \frac{3C_F}{4\pi}\alpha_s\right).\end{aligned} \tag{135}$$

Substituting the color Casimir $C_F = \frac{4}{3}$:

$$\sigma^{\text{NLO}}(e^+e^- \to \text{hadrons}) = \sigma_0 \left(1 + \frac{\alpha_s(s)}{\pi}\right). \tag{136}$$

3.7 Universal Factorization in the Singular Limits

In fixed-order perturbation theory, multi-parton amplitudes develop severe infrared (IR) divergences when additional emitted partons are soft (energy $E \to 0$) or collinear (angle $\theta \to 0$) relative to the emitting colored parton. Far from being merely computational hurdles, these singular limits have a universal, process-independent factorization structure that underpins subtraction methods, all-order resummation and the Monte Carlo parton shower paradigm.

3.7.1 The Soft Limit and Eikonal Currents

Consider the emission of a soft gluon with momentum k^μ ($k^0 = \omega \ll E_i$) from an outgoing on-shell quark line with momentum p^μ ($p^2 = m^2$). The uncontracted Feynman amplitude for this emission is given by:

$$\mathcal{M}_{n+1} \sim \bar{u}(p)\,(-ig_s T^a \not{\epsilon}^*(k))\,\frac{i(\not{p} + \not{k} + m)}{(p+k)^2 - m^2}\,\Gamma, \tag{137}$$

where Γ represents the remainder of the hard scattering process. Using the Dirac equation $\bar{u}(p)(\not{p} - m) = 0$ and the on-shell condition $k^2 = 0$, the propagator denominator reduces to $(p+k)^2 - m^2 = 2p \cdot k$. The numerator becomes:

$$\bar{u}(p)\not{\epsilon}^*(\not{p} + \not{k} + m) = \bar{u}(p)\left(2p \cdot \epsilon^* - (\not{p} - m)\not{\epsilon}^* + \not{\epsilon}^*\not{k}\right) \approx 2(p \cdot \epsilon^*)\,\bar{u}(p), \tag{138}$$

where the non-commuting term $\not{\epsilon}^*\not{k} \sim \mathcal{O}(\omega)$ is neglected relative to $p \cdot \epsilon^* \sim \mathcal{O}(E_p)$. Hence, the emission amplitude simplifies to the classical Eikonal approximation:

$$\mathcal{M}_{n+1} \approx g_s T^a \left(\frac{p \cdot \epsilon^*(k)}{p \cdot k}\right) \mathcal{M}_n. \tag{139}$$

Notice that the soft gluon emission is entirely spin-independent; the soft gluon couples exclusively to the classical color current associated with the 4-velocity of the emitting particle.

For an overall color-singlet dipole system (e.g., $e^+e^- \to q(p_1) + \bar{q}(p_2) + g(k)$), the amplitude is the sum of the radiation from both legs:

$$\mathcal{M}_{q\bar{q}g} \approx g_s T^a \left(\frac{p_1 \cdot \epsilon^*}{p_1 \cdot k} - \frac{p_2 \cdot \epsilon^*}{p_2 \cdot k}\right) \mathcal{M}_{q\bar{q}}. \tag{140}$$

Squaring the amplitude and summing over physical gluon polarizations $\sum_{\text{pol}} \epsilon^\mu \epsilon^{*\nu} = -g^{\mu\nu}$ (in a physical gauge):

$$|\mathcal{M}_{q\bar{q}g}|^2 \simeq g_s^2 C_F \left(-\frac{p_1}{p_1 \cdot k} + \frac{p_2}{p_2 \cdot k} \right)^2 |\mathcal{M}_{q\bar{q}}|^2 = g_s^2 C_F \left[\frac{2p_1 \cdot p_2}{(p_1 \cdot k)(p_2 \cdot k)} \right] |\mathcal{M}_{q\bar{q}}|^2. \quad (141)$$

The single-gluon Lorentz-invariant phase space is:

$$d\Phi_g = \frac{d^3\vec{k}}{(2\pi)^3 2\omega} = \frac{\omega \, d\omega \, d\cos\theta \, d\phi}{2(2\pi)^3}. \quad (142)$$

Thus, the soft emission probability factorizes into:

$$d\mathcal{P}_{\text{soft}} = \frac{\alpha_s C_F}{\pi} \frac{d\omega}{\omega} \frac{d\cos\theta \, d\phi}{2\pi} \left[\frac{1 - \cos\theta_{12}}{(1 - \cos\theta_{1k})(1 - \cos\theta_{2k})} \right]. \quad (143)$$

3.7.2 The Collinear Limit and Splitting Kernels

When two partons b and c become collinear ($\theta_{bc} \to 0$, virtuality $t = (p_b + p_c)^2 \to 0$), the matrix element factorizes into the n-parton Born matrix element and the universal Altarelli-Parisi splitting functions:

$$d\sigma_{n+1} \xrightarrow{\text{collinear}} d\sigma_n \times \frac{\alpha_s}{2\pi} \frac{dt}{t} \, dz \, \frac{d\phi}{2\pi} \, \hat{P}_{a \to bc}(z), \quad (144)$$

where z is the longitudinal energy fraction carried away by daughter parton b, and $(1-z)$ is carried by daughter parton c. The unregularized (real-emission) splitting kernels are:

$$\hat{P}_{q \to qg}(z) = C_F \left[\frac{1 + z^2}{1 - z} \right], \quad (145)$$

$$\hat{P}_{g \to q\bar{q}}(z) = T_F \left[z^2 + (1 - z)^2 \right], \quad (146)$$

$$\hat{P}_{g \to gg}(z) = 2C_A \left[\frac{z}{1 - z} + \frac{1 - z}{z} + z(1 - z) \right]. \quad (147)$$

In the overlapping soft-collinear limit ($z \to 1$ and $t \to 0$), we recover the double-logarithmic structure:

$$\frac{dt}{t} dz \hat{P}_{q \to qg}(z) \approx 2C_F \frac{dt}{t} \frac{dz}{1 - z} \sim 2C_F \frac{dk_\perp}{k_\perp} \frac{d\omega}{\omega}. \quad (148)$$

3.8 NLO Subtraction Methods

For exclusive observables (e.g., differential jet cross-sections with kinematic cuts), analytical phase-space integration is impossible. To compute $\sigma_{\text{obs}}^{\text{NLO}}$ numerically via Monte Carlo, one must isolate and cancel IR singularities at the integrand level using subtraction methods.

The general idea is as follows. Let J_n and J_{n+1} be the measurement functions defining the observable for n and $n+1$ partons. The general NLO cross section is written as:

$$\sigma^{\mathrm{NLO}} = \int d\Phi_{n+1} \left[d\sigma^R J_{n+1} - d\sigma^S J_n \right] + \int d\Phi_n \left[d\sigma^V + \int_1 d\sigma^S \right] J_n, \tag{149}$$

where $d\sigma^S$ is a local counterterm that matches the pointwise singular behavior of the real cross-section $d\sigma^R$ in all soft and collinear limits:

$$\lim_{\text{soft/collinear}} \left(d\sigma^R - d\sigma^S \right) = 0. \tag{150}$$

Consequently:

1. The first bracket $[d\sigma^R J_{n+1} - d\sigma^S J_n]$ is strictly finite in 4 dimensions ($\epsilon = 0$) and can be integrated numerically.

2. The integrated subtraction term $\int_1 d\sigma^S$ is integrated analytically in D dimensions over the 1-parton unresolved phase space, yielding explicit poles in $1/\epsilon^2$ and $1/\epsilon$ that cancel the virtual poles in $d\sigma^V$.

Schemes at NLO:

- Catani-Seymour (CS) Dipole Subtraction [6]
- Frixione-Kunszt-Signer (FKS) Subtraction [7]

3.9 Higher-order calculations

3.9.1 Structure of the cross section

An NNLO calculation for a $2 \to 2$ multi-boson process ($pp \to V_1 V_2$) receives contributions from three distinct phase-space sectors:

$$\sigma^{\mathrm{NNLO}} = \int_m \mathrm{d}\sigma^{\mathrm{VV}} + \int_{m+1} \mathrm{d}\sigma^{\mathrm{RV}} + \int_{m+2} \mathrm{d}\sigma^{\mathrm{RR}}, \tag{151}$$

where:

- **Double-Virtual** ($\mathrm{d}\sigma^{\mathrm{VV}}$): Involves 2-loop amplitudes interfering with Born plus squared 1-loop amplitudes:

$$\mathrm{d}\sigma^{\mathrm{VV}} \propto 2\,\mathrm{Re}\langle \mathcal{M}_0 | \mathcal{M}_2 \rangle] + |\mathcal{M}_1|^2. \tag{152}$$

  Contains infrared and ultraviolet poles up to ϵ^{-4} in dimensional regularization ($D = 4 - 2\epsilon$).

- **Real-Virtual** ($\mathrm{d}\sigma^{\mathrm{RV}}$): 1-loop corrections to single-real emission matrix elements ($q\bar{q} \to V_1V_2 + g$). Features overlapping infrared singularities up to ϵ^{-3}.
- **Double-Real** ($\mathrm{d}\sigma^{\mathrm{RR}}$): Tree-level amplitudes with two unresolved partonic emissions ($q\bar{q} \to V_1V_2 + gg$, $qg \to V_1V_2 + qg$, etc.), yielding singular phase-space integrations up to ϵ^{-4}.

3.9.2 Modern NNLO Subtraction Formalisms

Because of overlapping and nested double-unresolved infrared limits, isolating singularities requires advanced subtraction methods:

- **q_T-Subtraction (Slicing):** Exploits the transverse momentum q_T of the diboson system. Above a slicing parameter $q_{T,\mathrm{cut}}$, the process corresponds to an NLO calculation of Y + jet. In the limit $q_T \to 0$, the singular cross section is predicted analytically via Soft-Collinear Effective Theory (SCET):

$$\mathrm{d}\sigma^{\mathrm{NNLO}} = \mathcal{H}^{\mathrm{NNLO}} \otimes \mathrm{d}\sigma^{\mathrm{Born}} + \left[\mathrm{d}\sigma^{\mathrm{NLO}}_{Y+\mathrm{jet}} - \mathrm{d}\sigma^{\mathrm{CT}}\right]_{q_T > q_{T\mathrm{cut}}}. \tag{153}$$

- **N-Jettiness Subtraction ($\mathcal{T}_N$):** Slicing technique based on global N-jettiness event shapes.
- **Antenna Subtraction:** Fully local subtraction formalisms based on antenna functions or inclusive kinematic projections.
- **Sector-improved residue subtraction:** also a fully local scheme similar to FKS. Depending on implementation with numerical[8] or analytical integration of subtraction terms.
- **Projection-to-Born**

3.10 Infrared and Collinear (IRC) Safety

An observable $\mathcal{O}$ defined for an n-parton state $\mathcal{O}_n(p_1, \ldots, p_n)$ is **Infrared and Collinear Safe** if it is invariant under the emission of an infinitesimally soft parton or a collinear splitting:

$$\lim_{p_{n+1}\to 0} \mathcal{O}_{n+1}(p_1, \ldots, p_n, p_{n+1}) = \mathcal{O}_n(p_1, \ldots, p_n) \quad \text{(Soft Safety)}, \tag{154}$$

$$\lim_{p_i \| p_j} \mathcal{O}_{n+1}(\ldots, p_i, p_j, \ldots) = \mathcal{O}_n(\ldots, p_i + p_j, \ldots) \quad \text{(Collinear Safety)}. \tag{155}$$

Examples:

- *IRC Safe:* Total cross-sections, Jet rates defined via sequential recombination algorithms, Thrust $T = \max_{\vec{n}} \frac{\sum_i |\vec{p}_i \cdot \vec{n}|}{\sum_i |\vec{p}_i|}$.
- *IRC Unsafe:* Total hadron multiplicity, the energy of the single hardest hadron in a jet. Naive jet flavor definitions.

3.11 Sequential Recombination Jet Algorithms

To map multi-particle hadronic final states to underlying partonic kinematics, modern collider physics employs sequential recombination algorithms. The distance metrics for the k_t family are:

$$d_{ij} = \min(k_{t,i}^{2p}, k_{t,j}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \quad d_{iB} = k_{t,i}^{2p}, \tag{156}$$

where $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$, and R is the jet radius parameter. Depending on the power p, we have different behaviours:

- k_t algorithm ($p = 1$) clusters soft particles first; sensitive to soft pileup background.
- Cambridge/Aachen ($p = 0$) is a purely geometric angular clustering; ideal for jet substructure.
- anti-k_t algorithm ($p = -1$) clusters hard particles first; produces circular, conical jets at LHC.

3.12 All-order resummation & parton showers

Fixed-order calculations can achieve high precision for inclusive observables. Inclusive observables are somewhat loosely defined but usually refer to observables that are not sensitive to a scale hierarchy. Scale hierarchies can arise in various ways: from large ratios of physical scales, such as the vector mass and the center-of-mass energy; from phase-space cuts, for example a transverse-momentum cut on the QCD radiation; or from sensitivity to some non-perturbative state, such as heavy-flavor jets.

In those cases, large-logarithm corrections can appear and spoil perturbative convergence. This problem can be remedied by reordering the perturbative series, which assumes exponentiation of these logarithms and expands the exponent in $\alpha_s^n \ln^m$ (the powers depend on the problem at hand). We will see how this structure arises in the following section. While these methods give the possibility to describe multi-particle emissions

3.12.1 The Sudakov Form Factor

In hard collisions involving a high energy scale $Q \gg \Lambda_{\rm QCD}$, multiple successive collinear and soft gluon emissions occur. At order α_s^n, these produce large logarithmic terms of the form:

$$\alpha_s^n \ln^{2n}\left(\frac{Q^2}{Q_0^2}\right), \tag{157}$$

where $Q_0 \sim 1$ GeV is the non-perturbative cutoff. When $\alpha_s \ln^2(Q^2/Q_0^2) \sim \mathcal{O}(1)$, the standard fixed-order perturbative expansion breaks down and requires an all-orders resummation. Because successive emissions in the collinear limit are strongly ordered in virtuality:

$$Q^2 \gg t_1 \gg t_2 \gg \cdots \gg t_n \gg t_0, \tag{158}$$

the evolution from the hard scale Q^2 down to the infrared cutoff t_0 can be formulated as an iterative Markovian branching process.

The probability for a parton a to branch into daughters $b + c$ in an infinitesimal evolution step $[t, t - dt]$ is:

$$d\mathcal{P}_a(t) = \frac{dt}{t}\frac{\alpha_s(t)}{2\pi}\sum_{b,c}\int_{z_{\min}(t)}^{z_{\max}(t)} dz\, \hat{P}_{a\to bc}(z). \tag{159}$$

Derivation of the Sudakov Form Factor To implement the shower as a Monte Carlo event generator, we must compute the probability distribution for the next branching scale without assuming a fixed number of emissions.

Let $\Delta_a(t_1, t_2)$ denote the probability that parton a evolves from an initial scale t_1 down to a lower scale t_2 **without emitting any resolvable radiation** (the non-emission probability).

Using the chain rule of independent probabilities:

$$\Delta_a(t_1, t - dt) = \Delta_a(t_1, t) \times (1 - d\mathcal{P}_a(t))\,. \tag{160}$$

Rearranging into a differential equation:

$$\frac{d\Delta_a(t_1, t)}{dt} = \Delta_a(t_1, t)\frac{d\mathcal{P}_a(t)}{dt}. \tag{161}$$

With the boundary condition $\Delta_a(t_1, t_1) = 1$, direct integration from t_2 to t_1 yields the **Sudakov Form Factor**:

$$\Delta_a(t_1, t_2) = \exp\left(-\int_{t_2}^{t_1}\frac{dt'}{t'}\frac{\alpha_s(t')}{2\pi}\sum_{b,c}\int_{z_{\min}(t')}^{z_{\max}(t')} dz\, \hat{P}_{a\to bc}(z)\right). \tag{162}$$

Physical Properties of the Sudakov Form Factor:

1. **Unitarity & Virtual Corrections:**
   By the conservation of total probability, the sum of all resolvable real emission cross-sections and all virtual loop corrections is identically unity. The exponent of the Sudakov form factor corresponds exactly to the virtual loop diagrams plus unresolved soft/collinear real emissions resummed to all orders.

2. **Infrared Cutoffs:**
   The bounds $z_{\min}(t)$ and $z_{\max}(t) = 1 - z_{\min}(t)$ are determined by kinematic limits (e.g., transverse momentum cutoff $k_\perp^2 \approx z(1-z)t > k_{\perp,0}^2$), preventing the unphysical integration over unresolved soft or collinear poles.

3.12.2 Monte Carlo Generation Algorithm

The shower evolution proceeds iteratively:

1. Given a current scale t_{current}, generate a uniform random variable $r \sim U(0, 1]$.

2. Solve for the next branching scale t_{next} via the inversion condition:

$$\Delta_a(t_{\text{current}}, t_{\text{next}}) = r. \tag{163}$$

3. If $t_{\text{next}} \leq t_0 \sim 1$ GeV^2, no further perturbative branching occurs. Mark the partons as final (and pass them to the hadronization model).

4. If $t_{\text{next}} > t_0$:

   - Select the branching $a \to bc$ with probability proportional to $\int dz\, \hat{P}_{a\to bc}(z)$.
   - Sample z according to the probability density $\hat{P}_{a\to bc}(z)$.
   - Generate an azimuthal angle $\phi \in [0, 2\pi)$ uniformly.
   - Reconstruct the exact 4-momenta of daughters b and c, and recursively repeat the algorithm for each daughter starting from scale t_{next}.

3.12.3 Matrix Element Matching and Merging

Complementarity of Matrix Elements and Parton Showers:

Fixed-Order Matrix Elements	**Parton Shower**
Hard, well-separated partons	Collinear and soft partons
low, fixed multiplicity	High, variable ($N \gg 10$)
Exact loops & color interference	Leading/Next-to-Leading Log (LL/NLL)
Diverges in soft/collinear limits	Regularized via unitary Sudakov form factors

3.12.4 Matching (e.g., POWHEG, MC@NLO)

Matching combines an NLO fixed-order calculation with a parton shower for a specific process (e.g., $pp \to Z + X$) such that:

1. The total cross section is accurate to NLO.
2. The first hard emission reproduces the real-emission matrix element at high p_T.
3. The soft/collinear emissions are resummed by the shower at low p_T.
4. Double counting between the shower's $\mathcal{O}(\alpha_s)$ expansion and the fixed-order real emission is systematically subtracted.

To provide exclusive hadron-level simulations:

- NLO+PS (POWHEG / MC@NLO): Matches NLO matrix elements with Parton Showers (PS), avoiding double counting of single-real emission by subtracting the shower approximation.
- NNLO+PS (MiNNLO$_{\text{PS}}$ / Geneva): Achieves NNLO accuracy for inclusive diboson observables simultaneously with NLO accuracy for VV+jet and leading-logarithmic shower evolution.

3.12.5 Merging (e.g., CKKW-L, MLM)

Merging combines multiple tree-level matrix elements with different final-state parton multiplicities ($X + 0$ jets, $X + 1$ jet, $X + 2$ jets, ...) into an inclusive sample. A merging scale Q_{match} separates the phase space:

- Emissions above Q_{match} are generated according to exact tree-level Matrix Elements and reweighted by Sudakov weights.
- Emissions below Q_{match} are generated by the Parton Shower.

3.12.6 (Analytic) All-Order Resummation

The Low-q_T Breakdown of Fixed-Order QCD In multi-boson production ($pp \to VV + X$), when the transverse momentum of the diboson system $q_T \equiv |\vec{p}_{T,V_1} + \vec{p}_{T,V_2}| \ll M_{VV}$, fixed-order perturbative QCD fails. Multiple soft and collinear gluon emissions produce large logarithmic terms at order α_s^n:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}q_T^2} \sim \frac{\sigma_0}{q_T^2} \sum_{n=1}^{\infty} \left(\frac{\alpha_s}{2\pi}\right)^n \sum_{k=1}^{2n} c_{n,k} \ln^{k-1}\left(\frac{M_{VV}^2}{q_T^2}\right). \tag{164}$$

As $q_T \to 0$, these large logarithms diverge, rendering the fixed-order prediction unphysical (e.g., negative cross sections). Reliable predictions require summing these terms to all orders in α_s via the Collins-Soper-Sterman (CSS) or Soft-Collinear Effective Theory (SCET) formalisms.

The CSS Resummation Formalism In impact parameter space (b-space, conjugate to q_T), the resummed partonic differential cross section factorizes:

$$\frac{\mathrm{d}\hat{\sigma}}{\mathrm{d}q_T^2} = \frac{M_{VV}^2}{s}\int_0^\infty \mathrm{d}b\,\frac{b}{2}\,J_0(bq_T)\,\mathcal{W}_{ab}(b,M_{VV}) + \left[\frac{\mathrm{d}\hat{\sigma}}{\mathrm{d}q_T^2}\right]_{\text{finite}}, \tag{165}$$

where the all-order form factor $\mathcal{W}$ is given by:

$$\mathcal{W}_{ab}(b,M_{VV}) = \mathcal{H}_{ab}(M_{VV},\alpha_s)\exp\left[-\int_{b_0^2/b^2}^{M_{VV}^2}\frac{\mathrm{d}\mu^2}{\mu^2}\left(A(\alpha_s(\mu))\ln\frac{M_{VV}^2}{\mu^2} + B(\alpha_s(\mu))\right)\right], \tag{166}$$

with $b_0 = 2e^{-\gamma_E}$, cusp anomalous dimensions $A(\alpha_s) = \sum_n A^{(n)}\alpha_s^n$, and non-cusp terms $B(\alpha_s)$.

Derivation: Sudakov Form Factor in the Leading-Logarithmic (LL) Limit We evaluate the double-logarithmic Sudakov suppression in impact parameter b-space:

$$S_{\mathrm{LL}}(b,Q^2) = \int_{b_0^2/b^2}^{Q^2}\frac{\mathrm{d}\mu^2}{\mu^2}A^{(1)}\left(\frac{\alpha_s(\mu^2)}{2\pi}\right)\ln\left(\frac{Q^2}{\mu^2}\right), \tag{167}$$

where $Q = M_{VV}$, $A^{(1)} = 2C_F$, and the 1-loop running coupling satisfies:

$$\alpha_s(\mu^2) = \frac{\alpha_s(Q^2)}{1+\beta_0\alpha_s(Q^2)\ln(\mu^2/Q^2)}, \qquad \beta_0 = \frac{11C_A - 2n_f}{12\pi}. \tag{168}$$

Step 1: Variable Transformation. Define the logarithmic variable $\xi = \ln(Q^2/\mu^2)$ such that $\mathrm{d}\mu^2/\mu^2 = -\mathrm{d}\xi$. The integration limits transform as:

- Upper limit $\mu^2 = Q^2 \implies \xi = 0$.
- Lower limit $\mu^2 = b_0^2/b^2 \implies \xi_{\max} = \ln(Q^2b^2/b_0^2) \equiv L$.

Step 2: Integral Formulation with Running Coupling. Substituting the running coupling $\alpha_s(\mu^2) = \frac{\alpha_s(Q^2)}{1-\beta_0\alpha_s(Q^2)\xi}$:

$$S_{\mathrm{LL}}(b,Q^2) = \frac{C_F\alpha_s(Q^2)}{\pi}\int_0^L \mathrm{d}\xi\,\frac{\xi}{1-\beta_0\alpha_s(Q^2)\xi}. \tag{169}$$

Step 3: Analytical Integration. Let $\lambda = \beta_0\alpha_s(Q^2)L$. We perform substitution with $u = 1 - \beta_0\alpha_s(Q^2)\xi$:

$$\int_0^L \mathrm{d}\xi \frac{\xi}{1-\beta_0\alpha_s\xi} = \frac{1}{(\beta_0\alpha_s)^2}\int_{1-\lambda}^1 \mathrm{d}u \frac{1-u}{u} \tag{170}$$

$$= \frac{1}{(\beta_0\alpha_s)^2}\left[-\ln(1-\lambda) - \lambda\right]. \tag{171}$$

Step 4: LL Form Factor & Fixed-Coupling Expansion. The all-order LL exponent becomes:

$$S_{\mathrm{LL}}(b, Q^2) = -\frac{C_F}{\pi\beta_0^2\alpha_s(Q^2)}\left[\lambda + \ln(1-\lambda)\right]. \tag{172}$$

In the fixed-coupling limit ($\beta_0 \to 0$ or small λ), expanding $\ln(1-\lambda) \approx -\lambda - \frac{1}{2}\lambda^2$:

$$S_{\mathrm{LL}}^{\mathrm{fixed}}(b, Q^2) \approx \frac{C_F\alpha_s}{2\pi}L^2 = \frac{C_F\alpha_s}{2\pi}\ln^2\left(\frac{Q^2b^2}{b_0^2}\right). \tag{173}$$

The Sudakov form factor $\exp[-S(b, Q^2)]$ suppresses large b (low q_T), curing the infrared divergence and creating the characteristic Sudakov peak.

4 LHC phenomenology

The most important multi-boson processes at the LHC are the direct production of di-bosons, vector-boson fusion, and vector-boson scattering processes. This lecture focuses on di-boson production, but final states with three or more heavy bosons have been constrained, albeit with much lower precision.

The state-of-the-art for di-boson production is generally NNLO QCD in conjunction with NLO EW corrections for fully leptonic decay channels. This is for the fully off-shell processes as well as the stable inclusive production of boson pairs:

- WW: NNLO QCD [9, 10], NLO Loop-induced [11, 12], Higgs interference [13], NNLO+EW [14].
- ZZ: NNLO QCD [15, 16, 17, 18], NLO loop-induced [19, 20], Higgs interference [21, 13], NNLO+EW [14].
- WZ: NNLO QCD [22, 23], NNLO+EW [14].
- $Z\gamma$: NNLO QCD [24, 25, 26]
- $W\gamma$: NNLO QCD [25]

These fixed-order results have additionally matched with parton-showers in the MiNNLO (for example [27]) and Geneva matching approaches.

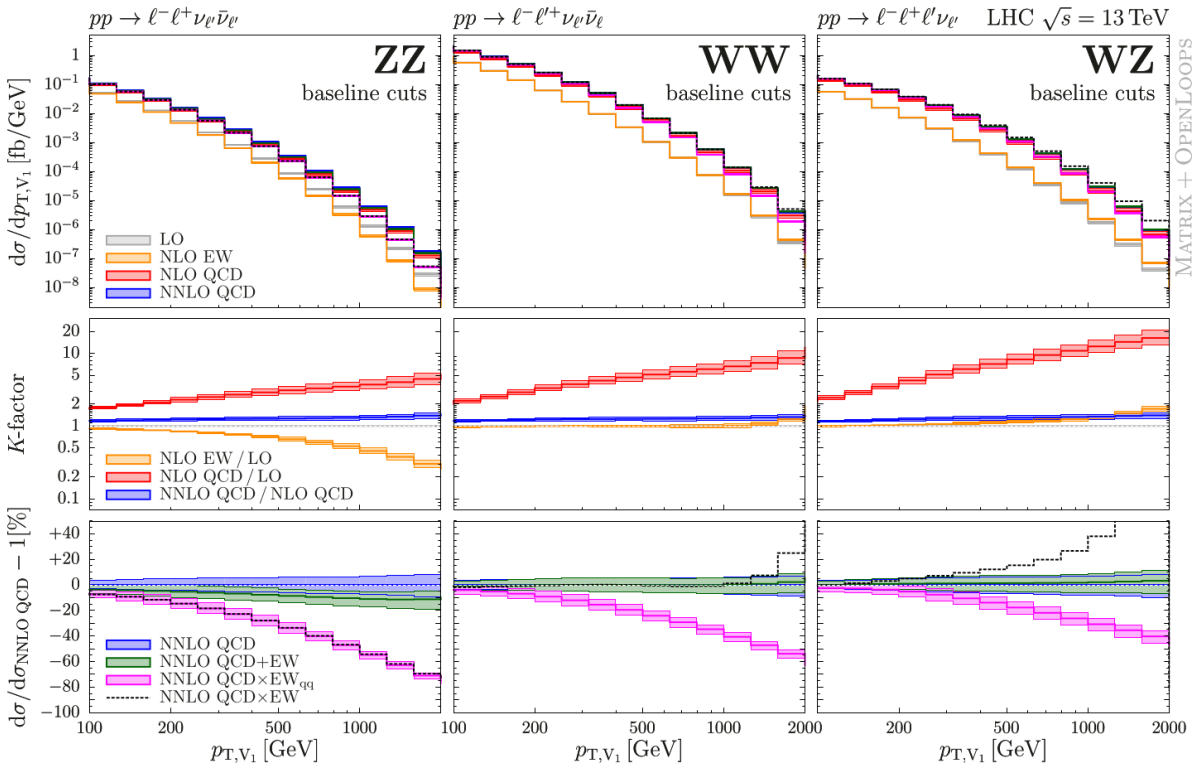


Figure 4: Transverse momentum spectra of the leading vector-boson. Plots taken from [14].

4.1 Higher-order QCD corrections for di-boson production processes

NLO QCD corrections for color-singlet production can be large at high transverse momenta. These 'giant k-factors' arise from implicit phase space constraints at LO. For example, consider a transverse momentum cut $p_T(V) > p_{T,cut}$ at LO. By momentum conservation, this implies $p_T(V') > p_{T,cut}$ at LO. At NLO, however, in the real corrections, the phase space $p_T(V') < p_{T,cut}$ can be filled through QCD radiation. Effectively, this results in an integration over the softer vector boson, yielding:

$$d\sigma^{V(V')j} \sim \sigma_{LO}^{Vj} \frac{\alpha_{EW}}{2\pi} \log^2(Q^2/m_V^2) \tag{174}$$

An example of this effect is shown in Figure 4. This effect is expected to be independent of the decay channel. The effects are really large in the multi-TeV region. This is beyond the reach for leptonic signatures but could be reached with hadronic ones.

Transitioning from NLO to Next-to-Next-to-Leading Order (NNLO) QCD reduces perturbative scale uncertainties from $\mathcal{O}(10-15\%)$ down to $\mathcal{O}(2-4\%)$, matching the experimental precision achieved by the ATLAS and CMS collaborations at the LHC.

4.1.1 Loop-Induced Gluon Fusion

Because gluons carry no electroweak charge, the gluon-initiated production of neutral gauge bosons ($gg \to \gamma\gamma, ZZ, W^+W^-, Z\gamma$) starts at 1-loop order via closed quark loops (box and triangle diagrams).

- **Formal Perturbative Order:** The amplitude is $\mathcal{O}(\alpha_s)$, so the squared matrix element is $\mathcal{O}(\alpha_s^2)$, making it formally an $\mathcal{O}(\alpha_s^2)$ NNLO QCD correction.
- **Gluon PDF Luminosity Enhancement:** Despite being loop-suppressed, the large gluon flux at small $x \sim 10^{-2}$–10^{-3} enhances this contribution, accounting for **10%–15%** of the total inclusive W^+W^- and ZZ cross sections at 13.6 TeV. WZ does not have any loop-induced contribution.
- **Higgs Signal-Background Interference:** Continuum $gg \to ZZ$ interferes destructively with $gg \to H^* \to ZZ$, providing a direct constraint on the off-shell Higgs boson width Γ_H.

4.2 NLO Electroweak (EW) Corrections

4.2.1 Electroweak Sudakov Logarithms

In the electroweak sector, singularity cancellation works slightly different from QCD. While QED is similar specifically W, Z exchanges/emissions behave differently:

1. Intermediate vector bosons possess physical masses ($M_W, M_Z > 0$), acting as physical infrared cutoffs, i.e., no soft/collinear limits. This holds for real emissions and virtual corrections. However, Integrations will yield logarithms of kinematic invariance and the regulator (boson mass): $\ln^n(s_{ij}/m_V^2)$.
2. Initial/final states carry explicit non-abelian $SU(2)_L$ weak charges. Incoming protons and isolated tagged leptons are not $SU(2)_L$ singlets, and real W/Z radiation produces physically distinguishable experimental signatures ($pp \to VV + W \neq pp \to VV$).

Consequently, virtual W and Z loops yield uncancelled **Electroweak Sudakov Logarithms** at center-of-mass energy squared $s \gg M_V^2$:

$$\delta_{\text{EW}}^{(1)} \sim -\frac{\alpha_{\text{EW}}}{4\pi \sin^2\theta_W} C_{\text{EW}} \ln^2\left(\frac{s}{M_V^2}\right) + \mathcal{O}\left(\ln\frac{s}{M_V^2}\right), \tag{175}$$

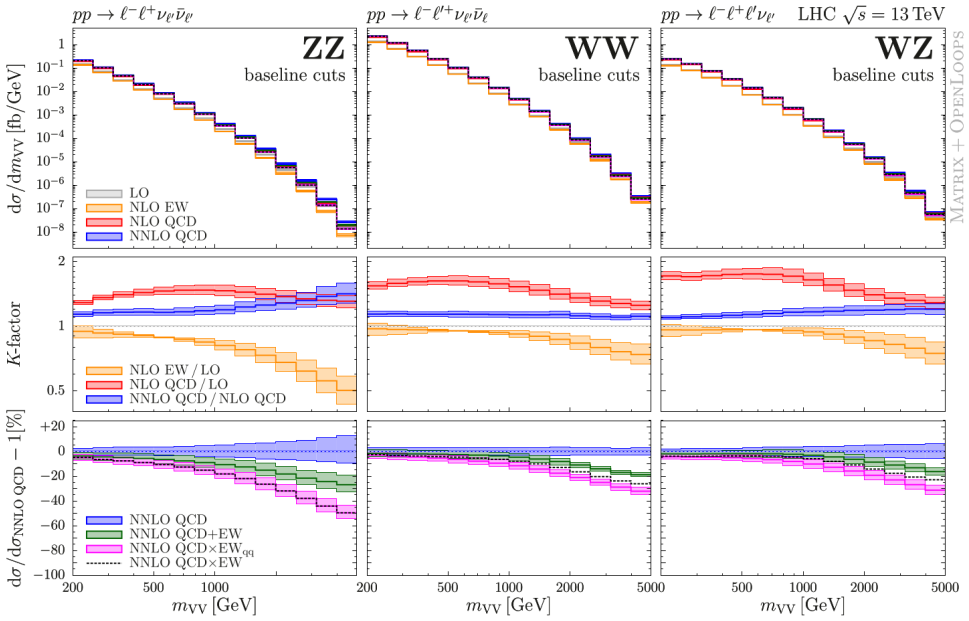


Figure 5: Diboson invariant mass distributions at different perturbative orders

where C_{EW} is the electroweak Casimir invariant. At high invariant masses ($m_{VV} \sim 2$ TeV) or large transverse momenta ($p_T^V \sim 1$ TeV), NLO EW corrections are **large and negative** (reaching -20% to -50%). This is demonstrated in Figure 4.

To understand that the EW Sudakov logarithms really depend on the kinematics, one can look at Figure 5, which shows the m_{VV} distribution. This is dominated by t- and u-channel exchanges (i.e., forward diboson), which are not particularly large and therefore have smaller corrections.

Derivation: Sudakov Double Logarithm from Massive Vector Exchange Consider the vertex correction for an on-shell fermion scattering off an external current with high momentum transfer $s = (p_1 - p_2)^2 \gg M_V^2$, where $p_1^2 = p_2^2 = 0$.

Step 1: Eikonal Approximation. The 1-loop massive vector exchange amplitude is given by:

$$\mathcal{M}_{\text{vert}} = -ig_V^2 \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{\bar{u}(p_2)\gamma^\alpha(\not{p}_2 - \not{k})\Gamma^\mu(\not{p}_1 - \not{k})\gamma_\alpha u(p_1)}{[k^2 - M_V^2 + i\delta][(p_1 - k)^2 + i\delta][(p_2 - k)^2 + i\delta]}. \tag{176}$$

In the soft limit ($k \ll p_1, p_2$), using the Dirac eikonal identities $(\not{p}_1 - \not{k})\gamma_\alpha u(p_1) \approx 2p_{1\alpha}u(p_1)$:

$$\mathcal{M}_{\text{vert}} \approx \mathcal{M}_{\text{Born}} \cdot \left[-ig_V^2 \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{4(p_1 \cdot p_2)}{[k^2 - M_V^2 + i\delta][k^2 - 2p_1 \cdot k + i\delta][k^2 - 2p_2 \cdot k + i\delta]}\right]. \tag{177}$$

Step 2: Sudakov Light-Cone Decomposition. Decompose loop momentum k^μ along light-like directions p_1 and p_2:

$$k^\mu = \alpha p_1^\mu + \beta p_2^\mu + k_\perp^\mu, \qquad \mathrm{d}^4k = \frac{s}{2}\,\mathrm{d}\alpha\,\mathrm{d}\beta\,\mathrm{d}^2k_\perp, \tag{178}$$

where $2(p_1 \cdot p_2) = s$, $k^2 = \alpha\beta s - |k_\perp|^2$. The scalar denominators approximate to:

$$k^2 - M_V^2 \approx -|k_\perp|^2 - M_V^2, \quad (p_1 - k)^2 \approx -\beta s, \quad (p_2 - k)^2 \approx -\alpha s. \tag{179}$$

Step 3: Longitudinal Integrations. Substituting into the loop integral:

$$I = \frac{1}{16\pi^2} \int \frac{\mathrm{d}|k_\perp|^2}{|k_\perp|^2 + M_V^2} \int \frac{\mathrm{d}\alpha}{\alpha} \int \frac{\mathrm{d}\beta}{\beta}. \tag{180}$$

Phase-space boundaries set by on-shell conditions are:

$$M_V^2 \leq |k_\perp|^2 \leq s, \qquad \frac{|k_\perp|^2}{s} \leq \alpha, \beta \leq 1. \tag{181}$$

Evaluating the longitudinal integrals:

$$\int_{|k_\perp|^2/s}^1 \frac{\mathrm{d}\alpha}{\alpha} = \ln\left(\frac{s}{|k_\perp|^2}\right). \tag{182}$$

Step 4: Transverse Integral & Final Logarithm. Substituting the longitudinal results into the transverse momentum integral:

$$I = \frac{1}{16\pi^2} \int_{M_V^2}^s \frac{\mathrm{d}|k_\perp|^2}{|k_\perp|^2} \ln\left(\frac{s}{|k_\perp|^2}\right). \tag{183}$$

Substituting $y = \ln(s/|k_\perp|^2)$ with $\mathrm{d}y = -\mathrm{d}|k_\perp|^2/|k_\perp|^2$:

$$I = \frac{1}{16\pi^2} \int_0^{\ln\left(s/M_V^2\right)} y\,\mathrm{d}y = \frac{1}{32\pi^2} \ln^2\left(\frac{s}{M_V^2}\right). \tag{184}$$

Including coupling prefactors, the leading relative 1-loop electroweak correction is:

$$\frac{\delta\sigma^{\mathrm{EW}}}{\sigma^{\mathrm{Born}}} \approx -\frac{\alpha_{\mathrm{EW}}}{2\pi \sin^2\theta_W} \ln^2\left(\frac{s}{M_V^2}\right). \tag{185}$$

4.2.2 Photon-Induced Sub-processes

NLO EW calculations require QED-evolved PDFs (e.g., LUXqed) containing a photon density $f_\gamma(x, \mu^2)$. Photon-photon scattering ($\gamma\gamma \to W^+W^-$) via t-channel W exchange produces large positive contributions at high m_{WW}, partially counteracting the negative EW Sudakov suppression. This is quite similar to the gg induced. But NLO EW also opens channels like γq as real emission contributions.

4.3 Comparison: NNLO QCD vs. NLO Electroweak

To summarize the impact of the different corrections:

NNLO QCD ($pp \to VV$)	**NLO EW** ($pp \to VV$)
$\mathcal{O}(\alpha_s^2) \sim (0.118)^2 \approx 1.4\%$	$\mathcal{O}(\alpha_{\text{EW}}) \sim 1/128 \approx 0.8\%$
Incl. corr. +10% to +20% (Large gg channel)	−3% to −5% (Moderate)
TeV Tails ($p_T > 1$ TeV) Modest increase	−30% to −50% (Sudakov suppression)

Eventually, we need to think on how to combine these predictions for the best accuracy. There different precriptions available [22, 23, 28, 14, 29, 30]:

$$\mathrm{d}\sigma_{NNLOQCD+EW} = \mathrm{d}\sigma_{LO}(1 + \delta_{QCD} + \delta_{EW} + \mathrm{d}\sigma_{LO}^{gg}) \tag{186}$$

$$\mathrm{d}\sigma_{NNLOQCD\times EW} = \mathrm{d}\sigma_{LO}(1 + \delta_{QCD})(1 + \delta_{EW} + \mathrm{d}\sigma_{LO}^{gg}) \tag{187}$$

The times correction can capture mixed QCD-EW effects if the dominant QCD and EW effects factorize, for example in the limit where Sudakov logarithms dominate.

These decompositions can be further refined by calculating the δ_{EW} and δ_{QCD} for individual initial state channels and applying either the multiplicative or additive description depending on whether the corrections are expected to factorize (then one can potentially achieve a better combination) or not.

4.4 Vector Boson Scattering (VBS)

Vector-boson scattering is the golden channel for investigations of the EWSB and has been extensively studied at the LHC (ATLAS [31, 32, 33], CMS [34, 35])

Signatures and Gauge-EW Topologies Vector Boson Scattering ($VV \to VV$) represents the cornerstone of electroweak symmetry breaking tests at the LHC:

$$pp \longrightarrow V_1V_2j_1j_2 + X. \tag{188}$$

Experimental identification relies on distinct VBS topology cuts:

- Two energetic forward/backward tagging jets with large dijet invariant mass ($m_{jj} >$ 500–1000 GeV).
- Large pseudorapidity separation between tag jets ($\Delta\eta_{jj} > 3.5$–4.5).
- A central rapidity gap (suppression of hadronic jet activity between the tagging jets due to colorless t-channel electroweak exchange).

Figure 6: Topology of electroweak $pp \to VVjj$ production at the LHC.

- **Multileg Matrix Element Complexity:** $pp \to 4\ell + 2j$ is an electroweak $2 \to 6$ process involving hundreds of Feynman diagrams; consistent finite-width modeling requires the Complex Mass Scheme to preserve gauge invariance.
- **NLO Electroweak Corrections:** At high invariant masses and transverse momenta ($p_T^V \gg M_V$), negative virtual EW corrections—dominated by double Sudakov logarithms $\propto -\alpha_{\text{EW}} \ln^2(s/M_W^2)$—suppress differential cross sections in the kinematic tails by -15% to -30%.
- **QCD Higher Orders:** QCD corrections stabilize the large renormalization and factorization scale uncertainties (μ_R, μ_F) inherent to dijet observables.
- **Resummation:** The central jet veto requires resummation. This is typically done via parton-showers. However, the complex topology is very sensitive to the various choices made in these simulations. Care has to be taken that parton-shower do not mess up the results.

Interference with QCD-Induced Multi-Boson Backgrounds The final state V_1 V_2 j_1 j_2 receives contributions from:

1. **Electroweak production ($\mathcal{O}(\alpha_{\mathbf{EW}}^6)$):** Pure VBS signal and non-VBS electroweak topologies.
2. **QCD-induced production ($\mathcal{O}(\alpha_s^2\alpha_{\mathbf{EW}}^4)$):** gluon exchange
3. **EW-QCD Interference ($\mathcal{O}(\alpha_s\alpha_{\mathbf{EW}}^5)$):** Non-negligible interference that must be accounted for in precision measurements.
4. Same-Sign $W^\pm W^\pm jj$: The golden channel for experimental measurement due to the absence of gluon-gluon initiated diagrams at leading order, drastically suppressing QCD backgrounds ($\mathcal{O}(\alpha_s^2\alpha_{\text{EW}}^4)$) relative to the electroweak signal ($\mathcal{O}(\alpha_{\text{EW}}^6)$).

Example measurements In figure 7 you can see to example differential measurements as a function of the di-jet invariant mass. You can observe how the signal (the electro weak process) becomes more and more dominated as the invariant mass grows.

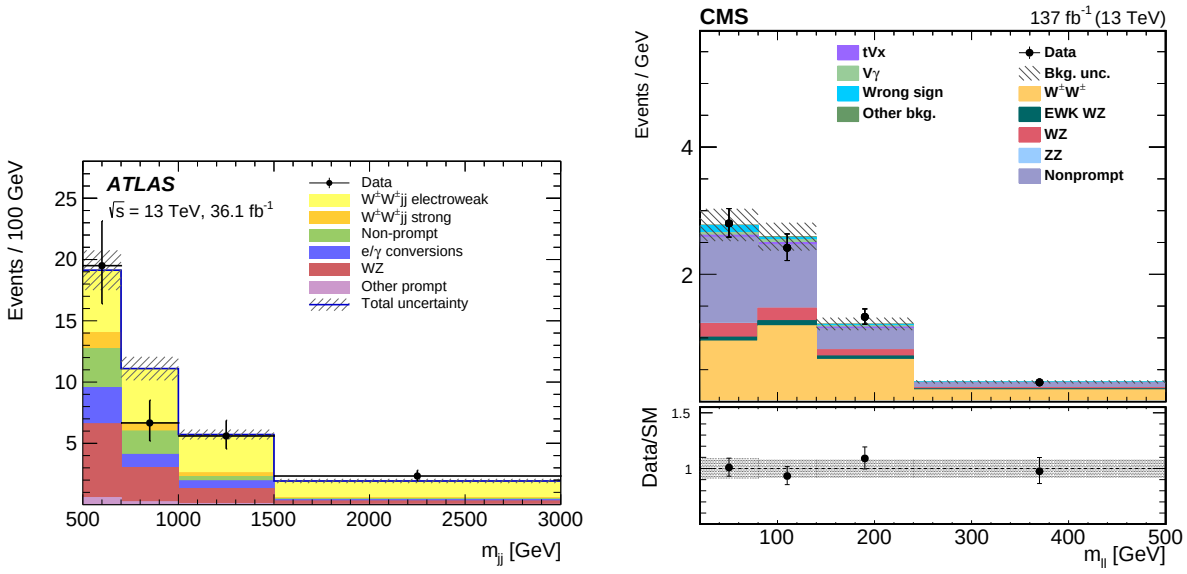


Figure 7: Example VBS measurements by ATLAS [32] and CMS [34].

5 Polarised states

The first question one should ask is why it is interesting to look at the polarization of vector bosons?

It is clear that the study of longitudinal bosons allows a link to the Goldstone bosons, EWSB, and the overall EW structureBut on the other side, and to clear the air: LHC collisions are unpolarised; we don't control the polarisation of the protons.s. We can't directly measure the polarisation of heavy particles either because they decay to fast.

However, the V-A structure of the electroweak interactions induces a strong correlation between the boson's polarisation (in a given frame of reference and direction) anits's decay products. This fact can be exploited to extract polarisation information fromunpolarisedd data.Polarisation fractions are pseudo observables as they will rely on the underlying theory modeling and choices (polarisation frame + definition, on-shell projection methods, perturbative accuracy, other approximations) we are using in this extraction..

The 'unpolarised' data contains all the information we have access to. Eventually you could think about a fully multi-differential measurement of all independent kinematic features of the final state. A comparison to theory of such a prediction is the ultimate test and would contain all information we would be interested in.

Why polarisation then? In a way, you can think about it as a way to increase the sensitivity to certain aspects of the EW theory by allowing us, within a given model, to correlate various observables in terms of a few 'free' (actually fully predicted in a given model) parameters.

Coeff.	$g_j(\theta^*, \phi^*)$	**Physical Origin**
A_0	$\frac{1}{2}(1 - 3\cos^2\theta^*)$	Longitudinal polarization fraction (f_0)
A_1	$\sin 2\theta^* \cos\phi^*$	Longitudinal–Transverse interference
A_2	$\frac{1}{2}\sin^2\theta^* \cos 2\phi^*$	Transverse Left–Right interference
A_3	$\sin\theta^* \cos\phi^*$	Parity-violating (axial-vector) interference
A_4	$\cos\theta^*$	Forward–Backward asymmetry / $V - A$ parity violation
A_5	$\sin^2\theta^* \sin 2\phi^*$	Vanishes at LO; CP / higher-loop sensitive
A_6	$\sin 2\theta^* \sin\phi^*$	Vanishes at LO; CP / higher-loop sensitive
A_7	$\sin\theta^* \sin\phi^*$	Vanishes at LO; CP / higher-loop sensitive

Table 1: The eight angular coefficients and their corresponding physical mechanisms.

5.1 Angular Coefficients Formalism

Consider the differential cross section for the production and decay of a single massive vector boson. Due to the 2-particle kinematics we can perform a decomposition in spherical harmonics [36, 37, 38]:

$$\frac{d\sigma}{d^4q\, d\cos\theta^*\, d\phi^*} = \frac{3}{16\pi}\frac{d\sigma}{d^4q}\left[(1+\cos^2\theta^*) + \frac{A_0}{2}(1-3\cos^2\theta^*) + \sum_{j=1}^{7} A_j g_j(\theta^*,\phi^*)\right], \quad (189)$$

where the decay angles (θ^*, ϕ^*) are defined in the chosen reference frame.

In Monte Carlo simulations, the angular coefficients $A_i(q)$ can be extracted as expectation values of orthogonal spherical harmonics:

$$\langle g_i(\theta^*,\phi^*)\rangle \equiv \frac{\int d\Omega\, g_i(\theta^*,\phi^*)\,\frac{d\sigma}{d\Omega}}{\int d\Omega\,\frac{d\sigma}{d\Omega}} \implies A_i = C_i\langle g_i(\theta^*,\phi^*)\rangle, \quad (190)$$

with the coefficients C_i:

$$A_0 = 4 - 10\langle\cos^2\theta^*\rangle, \quad C_1 = 5, \quad C_2 = 10, \quad C_3 = 4, \quad C_4 = 4. \quad (191)$$

The Lam-Tung Relation At $\mathcal{O}(\alpha_s)$ (LO for V + jet), rotational invariance and the spin-1/2 nature of quarks lead to the **Lam-Tung relation**:

$$A_0 - A_2 = 0. \quad (192)$$

This relation is broken only at NNLO ($\mathcal{O}(\alpha_s^2)$) by non-planar Feynman topologies and gluon-gluon initiated diagrams, making $A_0 - A_2$ a clean probe of multi-parton QCD dynamics.

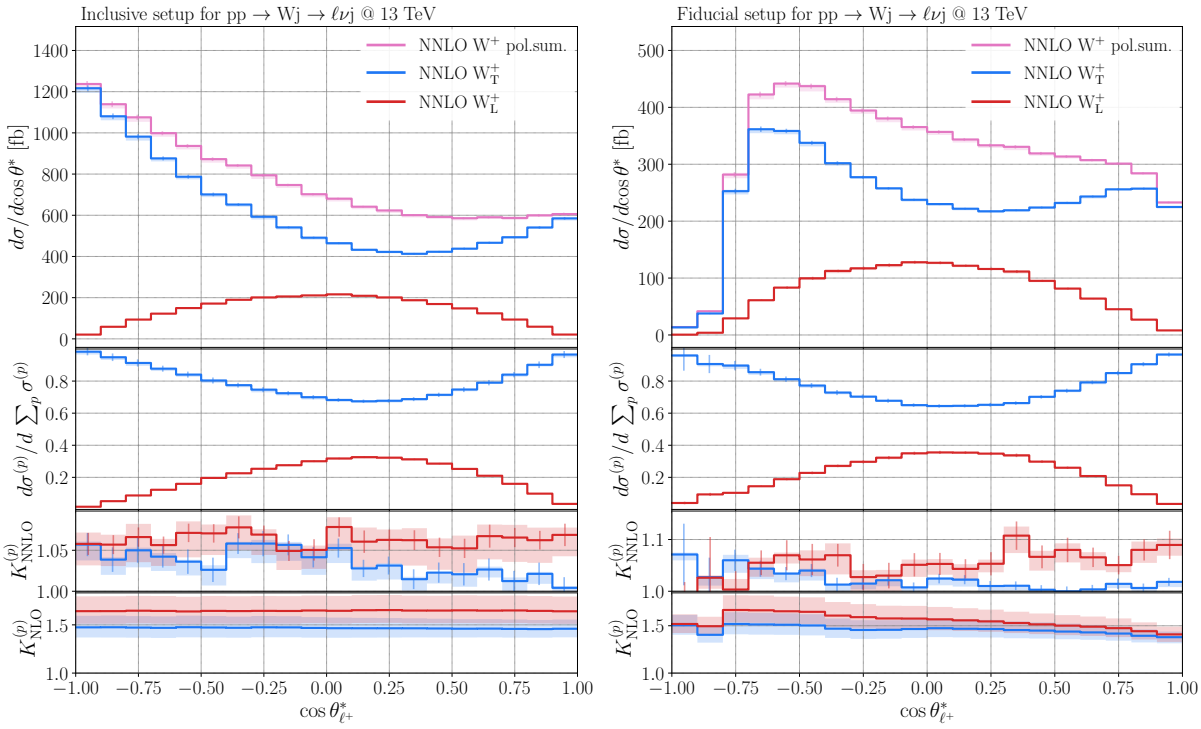


Figure 8: Demonstration of impact of fiducial phase space cuts on the $\cos\theta^*$

Limitations

- The integration assumes an inclusive phase space. Phase space cuts lead to artifacts 8.
- The decomposition is based on the 2-body decay. Photon radiation for example spoils this property.

An alternative can be provided by using template cross sections. The idea here is to make predictions for given propagating polarisations and to perform a fit of these to data.

5.2 Polarization Vectors for Massive Spin-1 Fields

For an on-shell vector boson with four-momentum $q^\mu = (E, \vec{q})$ such that $q^2 = M^2$, the polarization vectors $\varepsilon^\mu_\lambda(q)$ satisfy the orthonormality and transversality relations:

$$\varepsilon_\lambda(q) \cdot q = 0, \quad \varepsilon_\lambda(q) \cdot \varepsilon^*_{\lambda'}(q) = -\delta_{\lambda\lambda'}, \quad \lambda, \lambda' \in \{0, L, R\}. \tag{193}$$

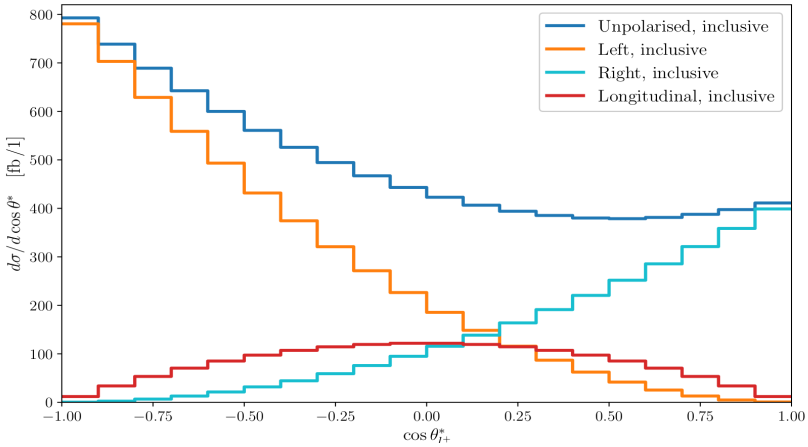


Figure 9: W^+ decay: angular dependence of the decay angle $\theta^*_{\ell^+}$ on the boson polarisation

Parametrizing the spatial momentum direction by spherical angles (θ, ϕ) in a given frame:

$$\varepsilon_L^\mu(q) = \frac{1}{\sqrt{2}}\,(0,\ \cos\theta\cos\phi + i\sin\phi,\ \cos\theta\sin\phi - i\cos\phi,\ -\sin\theta)\,, \tag{194}$$

$$\varepsilon_R^\mu(q) = \frac{1}{\sqrt{2}}\,(0,\ -\cos\theta\cos\phi + i\sin\phi,\ -\cos\theta\sin\phi - i\cos\phi,\ \sin\theta)\,, \tag{195}$$

$$\varepsilon_0^\mu(q) = \frac{1}{M}\,(|\vec{q}|,\ E\sin\theta\cos\phi,\ E\sin\theta\sin\phi,\ E\cos\theta)\,. \tag{196}$$

With this, we can compute, for example, polarised decay rates. You can see the W^+ decay in Figure 9. The strikingly different shapes are a key property we would like to exploit to extract the polarisation fractions from data.

5.2.1 Completeness Relation and Off-Shell Extension

On-shell, the polarization sum yields the numerator of the propagator, for example in unitary gauge:

$$\sum_{\lambda=0,L,R} \varepsilon_\lambda^{*\mu}(q)\varepsilon_\lambda^\nu(q) = -g^{\mu\nu} + \frac{q^\mu q^\nu}{M^2}. \tag{197}$$

For on-shell bosons, this connection is not accidental but is routed in the unitarity of the S-matrix. This fundamental connection does not exist for off-shell bosons. However, working in 4-dimensions, the relation can be reconstructed. Thus, for off-shell vector bosons ($q^2 \neq M^2$), a time-like (scalar) polarization state ε_t^μ is required to complete the four-dimensional basis:

$$\varepsilon_t^\mu(q) = \frac{\sqrt{q^2 - M^2}}{\sqrt{q^2}M} q^\mu \implies \sum_{\lambda=t,0,L,R} \varepsilon_\lambda^{*\mu}(q)\varepsilon_\lambda^\nu(q) = -g^{\mu\nu} + \frac{q^\mu q^\nu}{M^2}. \tag{198}$$

When decaying into massless conserved leptonic currents ($\mathcal{J}^\mu = \bar{u}\gamma^\mu v$ with $q_\mu \mathcal{J}^\mu = 0$), the contraction with ε_t^μ vanishes identically. The choice of the auxillary vectors (and for

that all other polarisation states in the off-shell case) is gauge dependent. This implies that any amplitude computed with off-shell polarisation vectors is gauge dependent.

For this lecture we will there constrain ourself to the case of on-shell definitions of polarisation.

5.2.2 Theoretical Choices

Because weak gauge bosons decay rapidly ($\Gamma_W \approx 2.085$ GeV), they only appear as intermediate states. Isolating a single polarized state naively from the complete $2 \to 4$ amplitude breaks $SU(2)_L \times U(1)_Y$ gauge invariance and drops quantum mechanical interference. Further considering a $2 \to 4$ process includes also diagrams that have no boson-propagator that can go onshell in the physical phase space and thus don't have a possible decomposition into polarisation states.

1. **Narrow-Width Approximation (NWA):** Formally takes $\Gamma_V/M_V \to 0$, factorizing production and decay:

$$\frac{1}{(q^2 - M^2)^2 + \Gamma^2 M^2} \xrightarrow{\text{NWA}} \frac{\pi}{M\Gamma}\delta(q^2 - M^2). \tag{199}$$

   Gauge invariance is strictly preserved, but off-shell effects and non-resonant background graphs are neglected.

2. **Double Pole Approximation (DPA):** Retains off-shell Breit-Wigner denominators while evaluating resonant matrix element numerators via an **On-Shell Projection (OSP)**:

$$\mathcal{M}_{\text{DPA}} = \sum_{\lambda_1,\lambda_2} \frac{\mathcal{A}^{\lambda_1\lambda_2}_{\text{prod}}(\hat{p}_i)\,\mathcal{A}^{\lambda_1}_{\text{dec}_1}(\hat{k}_j)\,\mathcal{A}^{\lambda_2}_{\text{dec}_2}(\hat{k}_l)}{(q_1^2 - M_1^2 + i\Gamma_1 M_1)(q_2^2 - M_2^2 + i\Gamma_2 M_2)}. \tag{200}$$

   The map $p \mapsto \hat{p}$ projects resonant four-momenta on-shell ($\hat{q}_i^2 = M_V^2$) while preserving total invariant mass and spatial orientations.

The impact of this choice is demonstrated in Figures 11 and 10.

The Complex Mass Scheme To describe unstable intermediate gauge bosons without violating Ward identities or gauge invariance, the **Complex Mass Scheme (CMS)** consistently promotes gauge boson masses to complex parameters throughout the Lagrangian:

$$\mu_W^2 = M_W^2 - iM_W\Gamma_W, \qquad \mu_Z^2 = M_Z^2 - iM_Z\Gamma_Z. \tag{201}$$

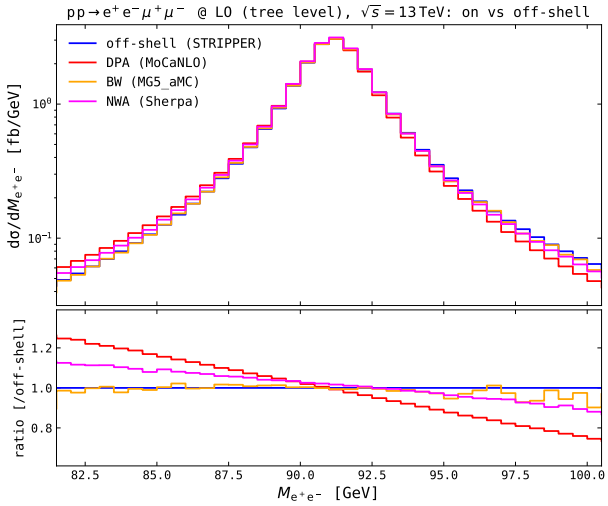


Figure 10: Differential distributions in the electron-positron invariant mass at tree level. The top panel compares absolute distributions obtained with the BW, NWA, and DPA approaches for the unpolarised process with the full off-shell result. The bottom panel shows the ratios over the full off-shell result. Plot from [39].

The weak mixing angle is defined analytically in the complex plane:

$$\cos^2\theta_W = \frac{\mu_W^2}{\mu_Z^2}, \qquad \sin^2\theta_W = 1 - \frac{\mu_W^2}{\mu_Z^2}. \tag{202}$$

Reference Frames Polarization fractions are frame-dependent. Two primary choices dominate collider studies:

- **Helicity Frame:** Defined in the rest frame of the decaying boson where the quantization axis $\hat{z}$ is aligned along the boson's direction of flight in the laboratory frame ($\vec{z} = \vec{q}_{\text{lab}}$). The $\hat{y}$-axis is perpendicular to the production plane.

- **Collins-Soper (CS) Frame:** The $\hat{z}$-axis bisects the angle between the two incoming hadron beam momenta $\vec{P}_1$ and $-\vec{P}_2$ boosted into the boson rest frame:

$$\hat{z}_{\text{CS}} = \frac{\vec{P}_1}{|\vec{P}_1|} - \frac{\vec{P}_2}{|\vec{P}_2|}. \tag{203}$$

  The CS frame minimizes sensitivity to initial-state QCD transverse momentum radiation (k_T).

- **Lab frame:** technically, simply defining everything in the laboratory frame is also possible. However, due to the enhanced sensitivity to recoil and boosts this choices is disfavored.

5.3 Polarised matrix elements

To define the cross section for fixed polarization states of the Z-bosons, we consider the following decomposition of the tree-level amplitude for the $ab \to e^+e^-\mu^+\mu^-$ process:

$$
\begin{aligned}
&\mathcal{M}^{ab\to e^+e^-\mu^+\mu^-} \\
&= \mathcal{M}^{ab\to Z_1Z_2}_{\mu_1\mu_2} \left[\prod_{k\in\{1,2\}} \frac{-g^{\mu_k}_{\nu_k}}{(p^2_{Z_k} - M^2_Z) + iM_Z\Gamma_Z}\right] \Gamma^{Z_1\to e^+e^-}_{\nu_1}\Gamma^{Z_2\to\mu^+\mu^-}_{\nu_2} \\
&= \mathcal{M}^{ab\to Z_1Z_2}_{\mu_1\mu_2} \left[\prod_{k\in\{1,2\}} \frac{-\sum_{\lambda_k}\epsilon^{\mu_k}_{\lambda_k}\epsilon^{\nu_k}_{\lambda_k}}{(p^2_{Z_k} - M^2_Z) + iM_Z\Gamma_Z}\right] \Gamma^{Z_1\to e^+e^-}_{\nu_1}\Gamma^{Z_2\to\mu^+\mu^-}_{\nu_2} \qquad (204) \\
&= \sum_{\lambda_1}\sum_{\lambda_2} \mathcal{M}^{ab\to Z_1Z_2}_{\lambda_1\lambda_2}\Gamma^{Z_1\to e^+e^-}_{\lambda_1}\Gamma^{Z_2\to\mu^+\mu^-}_{\lambda_2} \left[\prod_{k\in\{1,2\}} \frac{-1}{(p^2_{Z_k} - M^2_Z) + iM_Z\Gamma_Z}\right] \\
&\equiv \sum_{\lambda_1}\sum_{\lambda_2} \mathcal{M}^{ab\to Z_1Z_2\to e^+e^-\mu^+\mu^-}_{\lambda_1\lambda_2} \qquad (205)
\end{aligned}
$$

Here we identify Z_1 with the Z-boson that decays into electrons and Z_2 with the Z-boson that decays into muons. We keep the sum of the left and right polarization states coherent (that is, we keep the corresponding interference terms) and denote the corresponding transverse contribution to T. Using this decomposition, we can write the squared matrix elements that contribute to the cross section as a sum of polarized matrix elements and interference terms:

$$
\begin{aligned}
|\mathcal{M}|^2 &= \left(\sum_{\lambda_1,\lambda_2} \mathcal{M}^*_{\lambda_1,\lambda_2}\right)\left(\sum_{\lambda_1',\lambda_2'} \mathcal{M}_{\lambda_1',\lambda_2'}\right) \\
&= \underbrace{\sum_{\lambda_1,\lambda_2} |\mathcal{M}_{\lambda_1,\lambda_2}|^2}_{\text{pol. matrix elements}} + \underbrace{\sum_{\lambda_1,\lambda_2,\lambda_1',\lambda_2'} \mathcal{M}^*_{\lambda_1,\lambda_2}\mathcal{M}_{\lambda_1',\lambda_2'}\Big|_{\lambda_1\neq\lambda_1',\lambda_2\neq\lambda_2'}}_{\text{Interference}} \qquad (206)
\end{aligned}
$$

For the necessary matrix elements for QCD corrections, we can perform the same separation of polarization contributions for all higher-multiplicity and loop amplitudes because QCD corrections do not modify the Z propagators.

5.4 Polarised cross sections

Now, equipped with the knowledge on how to define polarised matrix elements, we can have a look at cross sections. Propagating the decomposed matrix elements to the cross-section level, we can write the cross-section as:

$$
\mathrm{d}\sigma = \sum_{\lambda,\lambda'\in\{\mathrm{T,L}\}} \mathrm{d}\sigma_{\lambda,\lambda'} + \text{int. terms.} \qquad (207)
$$

Considering perturbative corrections, we have an expansion in QCD for each $\mathrm{d}\sigma_{\lambda,\lambda'}$.

The interference terms can be computed by comparing the unpolarised but on-shell-projected cross section:

$$\mathrm{d}\sigma_{\mathrm{int}} = \mathrm{d}\sigma_{UU} - \sum_{\lambda,\lambda'\in\{\mathrm{T,L}\}} \mathrm{d}\sigma_{\lambda,\lambda'} \equiv \mathrm{d}\sigma_{UU} - \mathrm{d}\sigma_{SS} \tag{208}$$

where we introduced the unpolarized state 'U', meaning the propagator numerator has not been split, and $\mathrm{d}\sigma_{SS}$ as shorthand for the decoherent sum over the polarisations.

In addition, we will use the polarization fractions f_i for $i \in \{TT, TL, LT, LL\}$ defined by

$$\frac{\mathrm{d}\sigma_{SS}}{\sigma_{SS}} = \sum_{\lambda,\lambda'\in\{T,L\}} \frac{\sigma_{\lambda\lambda'}}{\sigma_{SS}} \frac{\mathrm{d}\sigma_{\lambda\lambda'}}{\sigma_{\lambda,\lambda'}} \equiv \sum_{\lambda,\lambda'\in\{T,L\}} \mu_{\lambda,\lambda'} \frac{\mathrm{d}\sigma_{\lambda\lambda'}}{\sigma_{\lambda,\lambda'}} \tag{209}$$

With this normalization, we have $\sum \mu_{\lambda,\lambda'} = 1$. A template fit would involve keeping these free-floating.

The off-shell effects, i.e., the error which is introduced through the OSP, can be assessed by comparing the off-shell computation and the unpolarized on-shell projected cross section:

$$\mathrm{d}\sigma_{\text{off. err.}} = \mathrm{d}\sigma_{\text{off}} - \mathrm{d}\sigma_{UU} \tag{210}$$

Using such techniques polarisation fractions have been measured by CMS [40, 41, 42] and ATLAS [43, 44, 45, 46, 47, 48].

5.5 Higher-order polarized diboson production

To ensure a good description of the data, we need to include large QCD and EW corrections for boson production processes. The literature has focused on di-boson production so far. At fixed order we have so far:

- W^+W^- polarisation NLO QCD [49], NLO EW [50, 51] NNLO QCD [52], top-quark effects [53]
- WZ NLO QCD+EW [54, 55, 56], EW [57], NNLO QCD + EW [58], semi-leptonic [59], QT [60]
- ZZ NLO QCD[61] NNLO QCD + NLO EW [39] LI [62]

At LO and NLO we also have parton-shower and multi-jet merged results:

- Madgraph LO + MJM [63]
- SHERPA nLO+PS [64]
- NLO+PS Powheg [65] with SMEFT [66]

Let's have a look at too examples.

5.5.1 Cross sections at NNLO QCD + NLO EW

In this section, we show fiducial results at fixed order for the ATLAS setup [45]. At NLO, we have QCD and EW corrections, which we decompose as:

$$\begin{aligned} \mathrm{d}\sigma^{\mathrm{QCD}}_{\mathrm{NLO}} &= \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{QCD},\,\mathrm{u\bar{d}}}\right)+\mathrm{d}\sigma_{qg} = \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{QCD}}\right), \\ \mathrm{d}\sigma^{\mathrm{EW}}_{\mathrm{NLO}} &= \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{EW},\,\mathrm{u\bar{d}}}\right)+\mathrm{d}\sigma_{q\gamma} = \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{EW}}\right). \end{aligned}$$

Similarly, we write the NNLO QCD corrections as

$$\begin{aligned} \mathrm{d}\sigma^{\mathrm{QCD}}_{\mathrm{NNLO}} &= \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{QCD},\,\mathrm{u\bar{d}}}+\delta^{(2)}_{\mathrm{QCD},\,\mathrm{u\bar{d}}}\right)+\mathrm{d}\sigma_{qg}\left(1+\delta^{(1)}_{\mathrm{QCD},\,qg}\right)+\mathrm{d}\sigma_{qq'}+\mathrm{d}\sigma_{\mathrm{gg}} \\ &= \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{QCD}}+\delta^{(2)}_{\mathrm{QCD}}\right). \end{aligned} \tag{211}$$

We then combine (N)NLO QCD and NLO EW corrections with both an additive and a multiplicative approach, according to:

$$\begin{aligned} \mathrm{d}\sigma^{(+)}_{\mathrm{NLO}} &= \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{QCD},\,\mathrm{u\bar{d}}}+\delta^{(1)}_{\mathrm{EW},\,\mathrm{u\bar{d}}}\right)+\mathrm{d}\sigma_{qg}+\mathrm{d}\sigma_{q\gamma}, \\ \mathrm{d}\sigma^{(\times)}_{\mathrm{NLO}} &= \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{QCD},\,\mathrm{u\bar{d}}}\right)\left(1+\delta^{(1)}_{\mathrm{EW},\,\mathrm{u\bar{d}}}\right)+\mathrm{d}\sigma_{qg}+\mathrm{d}\sigma_{q\gamma}, \\ \mathrm{d}\sigma^{(+)}_{\mathrm{NNLO}} &= \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{QCD},\,\mathrm{u\bar{d}}}+\delta^{(2)}_{\mathrm{QCD},\,\mathrm{u\bar{d}}}+\delta^{(1)}_{\mathrm{EW},\,\mathrm{u\bar{d}}}\right)+\mathrm{d}\sigma_{qg}\left(1+\delta^{(1)}_{\mathrm{QCD},\,qg}\right) \\ &\quad +\mathrm{d}\sigma_{qq'}+\mathrm{d}\sigma_{\mathrm{gg}}+\mathrm{d}\sigma_{q\gamma}, \\ \mathrm{d}\sigma^{(\times)}_{\mathrm{NNLO}} &= \mathrm{d}\sigma_{\mathrm{LO}}\left(1+\delta^{(1)}_{\mathrm{QCD},\,\mathrm{u\bar{d}}}+\delta^{(2)}_{\mathrm{QCD},\,\mathrm{u\bar{d}}}\right)\left(1+\delta^{(1)}_{\mathrm{EW},\,\mathrm{u\bar{d}}}\right)+\mathrm{d}\sigma_{qg}\left(1+\delta^{(1)}_{\mathrm{QCD},\,qg}\right) \\ &\quad +\mathrm{d}\sigma_{qq'}+\mathrm{d}\sigma_{\mathrm{gg}}+\mathrm{d}\sigma_{q\gamma}\,. \end{aligned} \tag{212}$$

5.5.2 COMETA study: differential cross section

In a study organized by COMETA, we studied in detail differences between various orders in perturbation theory and between different parton-shower matching/merging computations[39]. For this we used ZZ production.

An example result can be seen in 12.

	σ_{LO} (fb)	$\delta^{(1)}_{\text{QCD}}$	$\delta^{(2)}_{\text{QCD}}$	$\delta^{(1)}_{\text{EW}}$	$\sigma^{(+)}_{\text{NLO}}$ (fb)	$\sigma^{(\times)}_{\text{NLO}}$ (fb)	$\sigma^{(+)}_{\text{NNLO}}$ (fb)	$\sigma^{(\times)}_{\text{NNLO}}$ (fb)
off.	$19.869(2)^{+4.6\%}_{-5.7\%}$	+78.6%	+16.4%	−4.4%	$34.60(1)^{+5.4\%}_{-4.3\%}$	$34.19(1)^{+5.3\%}_{-4.2\%}$	$37.87(13)^{+2.5\%}_{-2.2\%}$	$37.50(13)^{+2.7\%}_{-2.2\%}$
unp.	$19.457(1)^{+4.6\%}_{-5.7\%}$	+79.0%	+16.3%	−4.4%	$33.957(6)^{+5.4\%}_{-4.3\%}$	$33.550(6)^{+5.3\%}_{-4.3\%}$	$37.13(8)^{+2.6\%}_{-2.2\%}$	$36.77(8)^{+2.7\%}_{-2.3\%}$
LL	$1.5326(1)^{+5.0\%}_{-6.1\%}$	+29.8%	+11.1%	−4.5%	$1.921(0)^{+2.8\%}_{-2.3\%}$	$1.895(0)^{+2.7\%}_{-2.2\%}$	$2.091(3)^{+2.1\%}_{-1.8\%}$	$2.064(3)^{+2.4\%}_{-1.7\%}$
LT	$2.0716(2)^{+5.6\%}_{-6.8\%}$	+159.2%	+42.7%	−3.5%	$5.295(1)^{+7.4\%}_{-5.9\%}$	$5.244(1)^{+7.3\%}_{-5.9\%}$	$6.181(23)^{+4.2\%}_{-3.4\%}$	$6.134(23)^{+4.3\%}_{-3.5\%}$
TL	$1.9531(4)^{+5.6\%}_{-6.8\%}$	+161.8%	+43.7%	−0.9%	$5.097(1)^{+7.4\%}_{-5.9\%}$	$5.046(1)^{+7.3\%}_{-5.9\%}$	$5.951(13)^{+3.9\%}_{-3.3\%}$	$5.940(13)^{+3.9\%}_{-3.3\%}$
TT	$13.751(1)^{+4.3\%}_{-5.4\%}$	+60.9%	+9.0%	−5.1%	$21.431(5)^{+4.6\%}_{-3.7\%}$	$21.167(5)^{+4.5\%}_{-3.7\%}$	$22.67(8)^{+1.6\%}_{-1.5\%}$	$22.43(8)^{+1.8\%}_{-1.6\%}$

Table 2: Fiducial cross sections (in fb) in the ATLAS setup [45]. The percentages δ's are NLO (1) and NNLO corrections (2), relative to the LO. The cross sections with $+(\times)$ labels understand an additive (multiplicative) combination of QCD and EW corrections, according to Eq. 212. For doubly polarised states, the first (second) label corresponds to the W(Z) boson. Numbers in parentheses represent MC-integration numerical uncertainties. Taken from [58].

Role of the Loop-Induced Contribution Although formally entering at $\mathcal{O}(\alpha_s^2)$ (NNLO), the gluon-fusion loop channel

$$gg \to W^+W^-, \quad gg \to ZZ \tag{213}$$

is enhanced by the large gluon parton luminosity at high energies and thus contributes significantly to the total scale variation band of the diboson cross section.

5.6 Higher-order uncertainties and polarisation fits

Consider the hard scattering process $pp \to W^{\pm}j \to \ell\nu j$ at $\sqrt{s} = 13$ TeV.

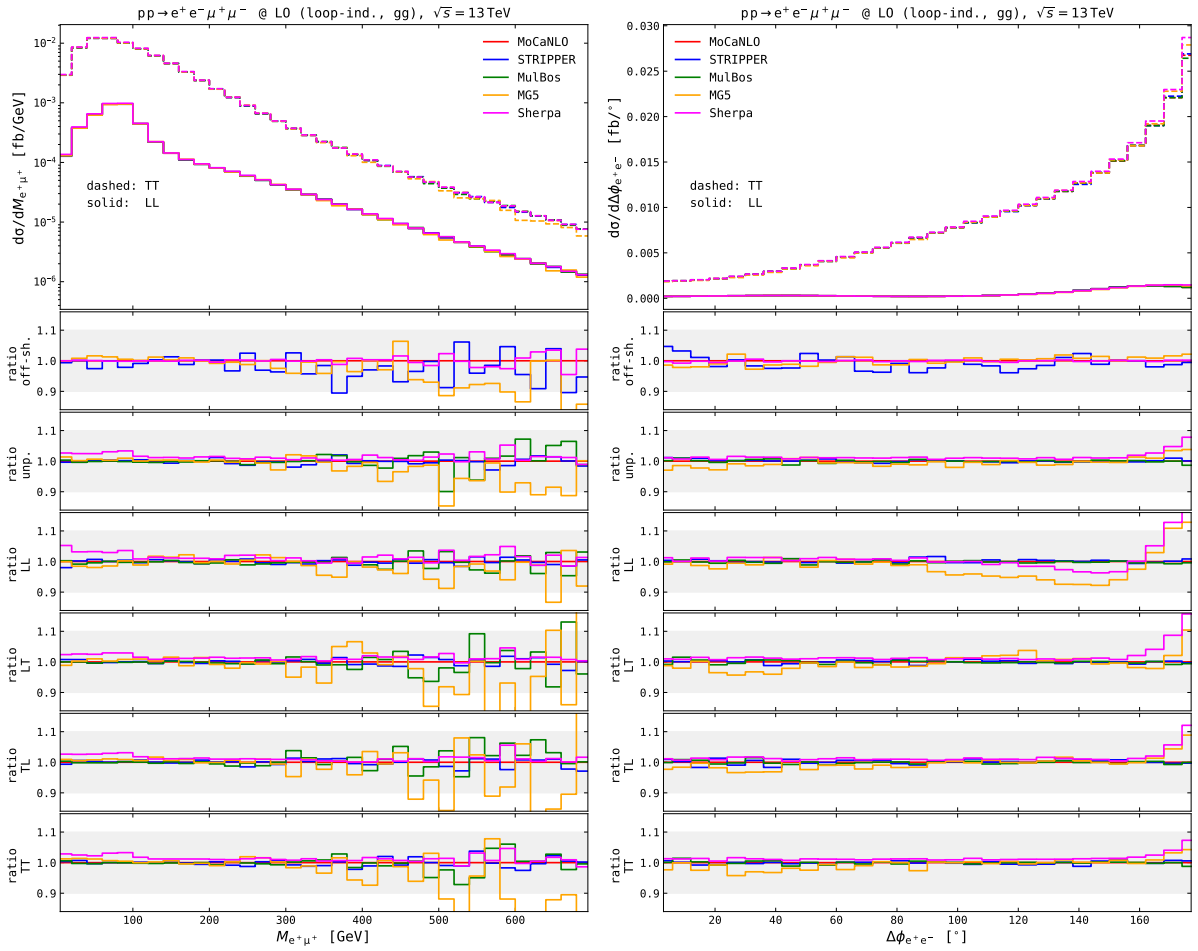
1. **Scale Uncertainties:** NLO scale bands are sizable ($\sim$ 15–25%) due to the opening of new gluon-initiated channels ($gq \to Wq'g$). At NNLO QCD, residual scale uncertainties shrink dramatically to the 1–2% level.

2. **Template Fitting Strategy:** Extracting polarization fractions from experimental data requires fitting calculated polarized shape templates σ_λ to unfolded $\cos\theta^*$ distributions:
$$\frac{d\sigma_{\text{unpol}}}{d\cos\theta^*} = f_L \left(\frac{d\sigma_L}{d\cos\theta^*}\right) + f_R \left(\frac{d\sigma_R}{d\cos\theta^*}\right) + f_0 \left(\frac{d\sigma_0}{d\cos\theta^*}\right). \tag{214}$$
Using NNLO templates significantly reduces fit biases that were previously present when using LO/NLO shapes.

Calculated within the Sector-Improved Residue Subtraction (STRIPPER) framework [67].

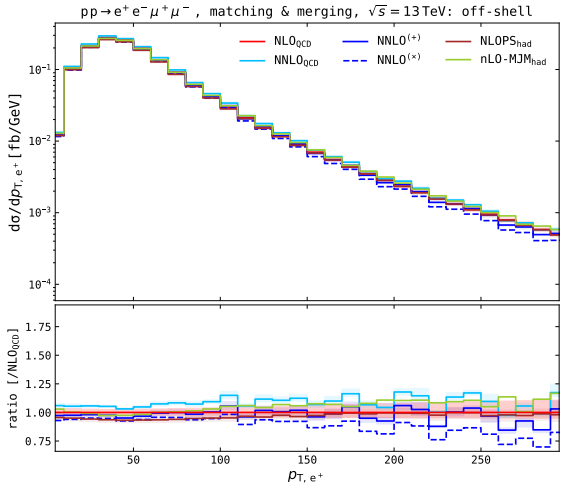
Acknowledgements

I would like to acknowledge the inspiration and input from various lectures on precision phenomenology at HEP schools, as well as standard QCD textbooks (citations in the main text). In particular, I build on the excellent QCD lectures by Giulia Zanderighi, Nigel Glover, and Gavin Salam, as well as Steffen Schumann's SM lecture. Particular thanks also go to Andrei Popescu, whose PhD thesis I used as the basis for the introduction to polarised states.

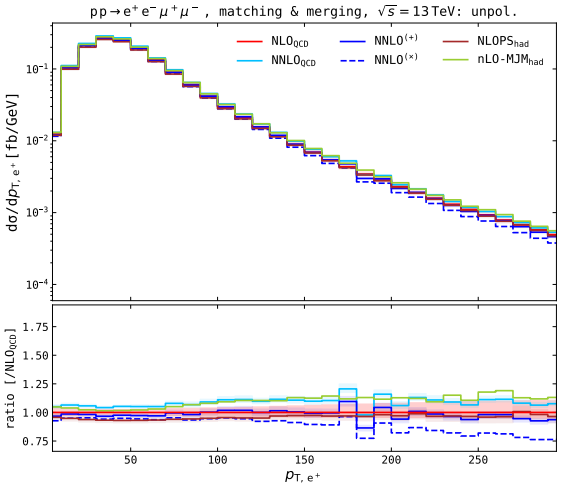


(a) Positron–antimuon invariant mass (b) Positron–electron azimuthal-angle separation

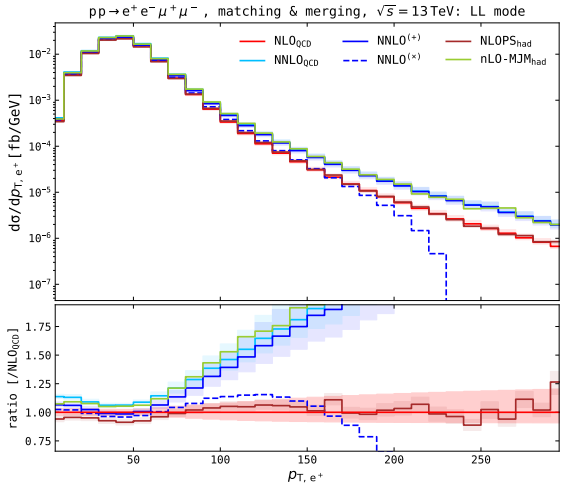
Figure 11: Differential distributions for the loop-induced gg contributions. Absolute distributions for the LL (solid curves) and TT (dashed curves) polarisation modes are shown in top panels. Ratio plots for the various modes (off-shell, unpolarised, LL, LT, TL, TT) are shown in the lower panels, relative to Mocanlo results. Shaded gray bands in the lower panels span ratios between 0.9 and 1.1. Plot from [39].



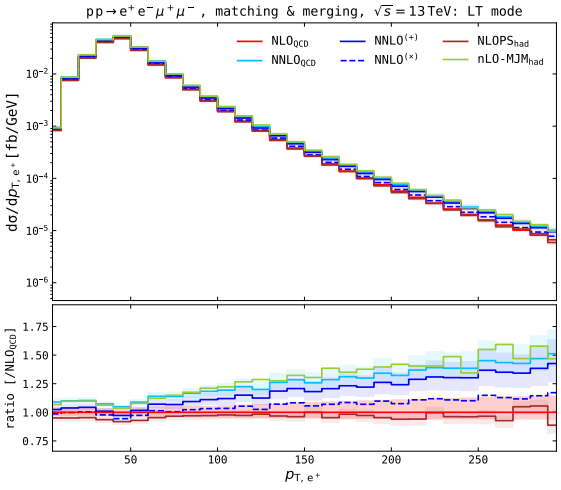
(a) off-shell



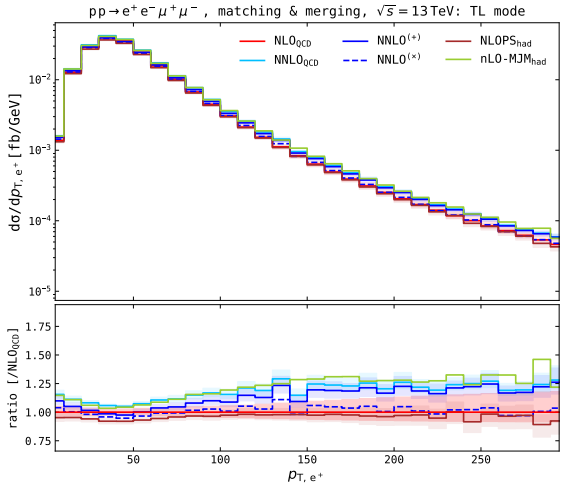
(b) unpol.



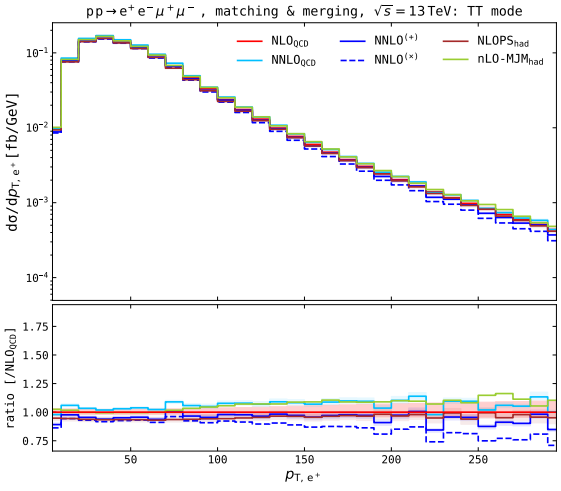
(c) LL



(d) LT



(e) TL



(f) TT

Figure 12: Fiducial distributions in the positron transverse momentum.

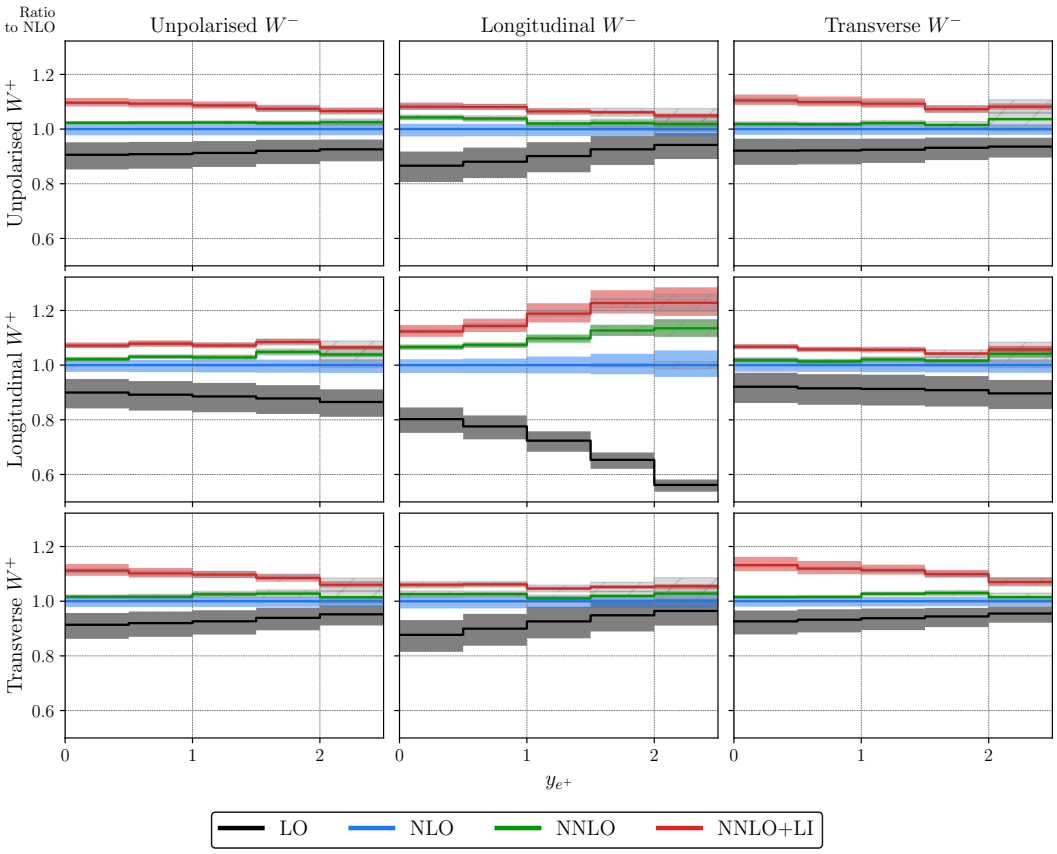


Figure 13: Impact of loop-induced contributions

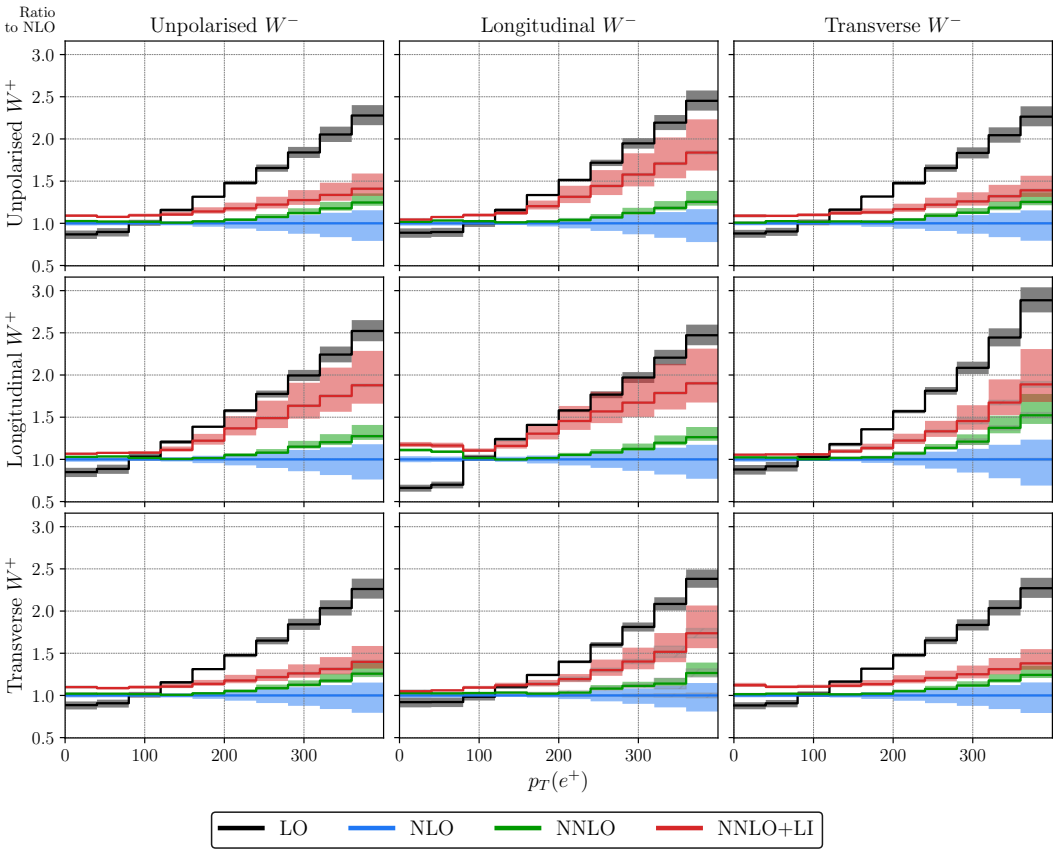


Figure 14: Impact of loop-induced contributions on the transverse energy of the positron.

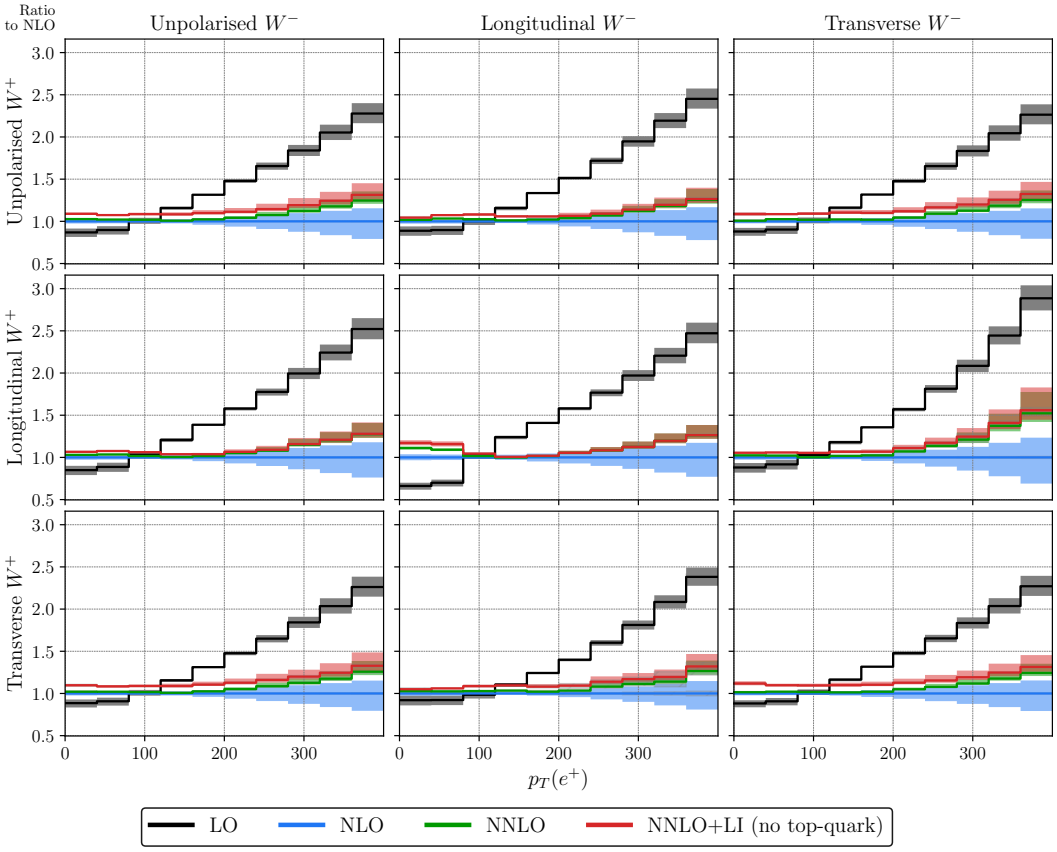


Figure 15: Impact of loop-induced contributions on the transverse energy of the positron. No top-quark loops

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